

Provenance

The Eugene A. Cernan Space Collection

APOLLO 17	
LM TIMELINE BOOK	
PART NO.	S/N
SKB32100123-388	1001

Flown to
Lunar
Module
XVII
E. Cernan

APOLLO 17

LM TIMELINE BOOK

NOVEMBER 7, 1972

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Landed on the Moon
aboard the Apollo 17 LM "Challenger"

Gene Cerny

ACKNOWLEDGMENT

<u>AREA</u>	<u>NAME/BRANCH</u>	<u>LOCATION</u>
Rendezvous (Sec. 2, 4)	S. Grega Flight Procedures CG4	Ext 5348 Bldg 4, Rm 252
Post Docking (Sec. 3)	G. Doerre Spacecraft Systems CG2	Ext 4371 Bldg 4, Rm 250

It is requested that any organization having comments in the above areas contact the responsible individuals.

THIS IS THE BEGINNING, UNDOCK TO DPS THROTTLE CHECK

NOT THE END

110:15

PREP FOR UNDOCKING

USE ACTIVATION & C.O.

C/L TO 10 MIN BEFORE UNDOCK

CHECK ATT (0, 180/284, 060)

V48 22012

LM WT 36711 (36,744)

PRO, V34

V06N20 COPY LM AND CSM ANGLES AND TIME

P47 0 - 110

UNDOCK & SEPARATION 110:24:00 (110:28) 110:27

V77E

P00, V60 TRIM TO .1 FPS

YAW LT 60°

PITCH UP 90°

*SEQUENCE CAMERA - ON (1 MIN) *

FDAI (0, 280/013, 0)

*VHF ANT - FWD (✓ LS VHF ANT DPLY) *

*SEQUENCE CAMERA - OFF *

HELMETS & GLOVES - OFF (OPT) *

*SUIT GAS DIVERTER - EGRESS *

*CABIN GAS RETURN - EGRESS *

LANDING RADAR CHECKOUT

CB(11) PGNS: LDG RDR - CLOSE

CK TEMP (60° - 95°)

X-PNTRS - HI MULT

MODE SEL - LR

TM SW - H/H

LDG ANT - AUTO

RDR TEST - LDG

POWER SIGNAL LIGHT OUT

TEST MON - ALT/VEL XMTR (2.1 - 5.0), AGC

X-PNTRS PEGGED UP, LEFT

TM - H (8000 ± 100), H (-480 ± 2)

V63 N12 OPT 2, PRO

N66 8286 ± 10, ANT POS 1 (00001), PRO

N67 V_x (-00495 ± 2), V_y (+01862 ± 2),

V_z (+01331 ± 2)

V34, RDR TEST OFF (ALT - 0, POWER SIGNAL

LIGHT ON, X-PNTRS - CENTERED)

CB(11) PGNS: LDG RDR - OPEN

CSM 00035

10467

00052

22012

110:35 KM

30109

28453

35948

AOS

TTO

+41

DPS THROTTLE CHECK

*CB(16) STAB/CONT: ENG ARM - CLOSE *

THROT CONT - MAN/CDR

TTCA (BOTH) - THROTTLE (MIN)

*VERIFY MSFN CONTACT *

ENG STOP - PUSH

ENG ARM - DES (DES REG LT - ON)

TTCA MIN (6.6% - 13.4%)

THEN SOFT STOP (46.2% - 59.2%)

THEN MAX (93.6% - 100+%)

THEN MIN

ADJUST FRICTION

MAN THROT - LMP

*REPEAT TEST FOR LMP TTCA *

ENG ARM - OFF

*CYCLE CWEA (DES REG LT - OFF) *

ENG STOP - RESET

THROT CONT - AUTO/CDR

TTCA (BOTH) - JETS

110:35

110:45

Landed on the Moon

aboard the Apollo 17 LM "Challenger"

Gene Cerny

DATE 10/18/72

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UNDOCK TO DPS
THROTTLE CHECK

110:45

DPS PRESS TO MSFN UPDATE

DPS PRESS & C.O.

22012

110:50

PRPLNT TEMP/PRESS MON - DES 1 & 2
FUEL 50°-75°F 50-130 PSI
OXID 50°-75°F 30-80 PSI
HELIUM MON: SUPCRIT PRESS 1070-1570 *1260*
 : AMB PRESS 1495-1750 *1620*
DES He REG 1 tb - GRAY, REG 2 tb - bp
MASTER ARM - ON
DES PRPLNT ISOL VLV - FIRE
He PRESS/DES START - FIRE
MASTER ARM - OFF
PRPLNT TEMP/PRESS MON: DES 1 & 2
FUEL & OXID 50°-90°F 200-250 PSI
HELIUM MON: AMB PRESS 200-1110
 : SUPCRIT PRESS 1070-1570

AGS ACTIVATION

*AGS STATUS - STBY (MASTER ALARM, *
* & AGS WARNING LT - ON) *
*CB(16) STAB/CONT: AEA - CLOSE *
* (AGS WARNING LT - OFF) *
CB(11) AC BUS B: AGS - CLOSE
*RECORD TIME *110:52:00*
*AGS STATUS - OPERATE (MASTER ALARM, *
* & AGS WARNING LT - ON) *
*O2/H2O QTY MON - C/W RESET *
*ATT MON (LMP) - AGS *
*412R+1 SELF TEST SATISFACTORY *
*000 +888888 (OPR ERR LT - ON) *
*123 -45679 (DO NOT ENTR) *
*V16 N65E *
*SET AGS TIME USING 110 HR BIAS *
* 377 _____ (+00450) *

*616+0

*224 _____ (+60457) *

*225 _____ (+29364) *

*226 _____ (+60366) *

*305 _____ (+00642) *

*662 _____ (-33007) *

*673 _____ (-54456) *

*574R DESCENT STAGE (+ NOT STAGED) *

*604R LUNAR SURFACE FLAG *

* (+ NOT ON LUNAR SURFACE) *

*612R STAGING COUNTER (+0 NOM) *

*232R +00600 *

*233R +00250 *

*464R +00500 *

*465R +00195 *

*623R +00000 *

*514R -60000 *

*515R -44223 *

*516R +00000 *

MSFN UPDATE

*COPY AGS K FACTOR *109:59:59.94*

V47E

V25E LOAD AGS K FACTOR UPDATE

*414+1 *

*400+3 (AFTER 50 16) *

V83, SET ORDEAL

*317R, 440R (*0.20NM, 1fps*) *

110:50

110:55

110:55

AGS CHECK TO COAS CAL

AGS CONT CHECK

MODE CONT (AGS) - ATT HOLD

GUID CONT - AGS

MNVN TO FDAI (0, 315/330, 0) IF DYNAMICS ABNORMAL, SAY "SHIT"

*CAMERA SETTINGS FOR REV 12 TCA

*LM3/DAC/10/CEX (T/2.8, 1/250, ∞)

* 1 FPS, .05 MAG 0, (5 MIN)

*LM/DC/60/CEX (f5.6, 1/125, 74 ft)

* 5 FR, MAG A

*SEQUENCE CAMERA - ON (110:56)

REV 12 LS TCA ____:____:____ (111:01)

*SEQUENCE CAMERA - OFF

RENDEZVOUS RADAR CHECKOUT

GUID CONT - PGNS

CB(11) AC BUS A: RNDZ RDR - CLOSE

CB(11) PGNS: RNDZ RDR - CLOSE

*VHF A XMTR - VOICE/RNG

✓ TEMP (10° - 75°)

RT/ERR MON - RR

RR SLEW, MANUAL LOCK-ON, RR LGC

TM - RNG/RNG RT

V63, PRO, NO TRACK LIGHT OUT, PRO, N78

COMPARE N78, VHF, TM
V34

*VHF A XMTR - VOICE *

V41N72E (+18000, +28300)

CB(11) PGNS: RNDZ RDR - OPEN

CB(11) AC BUS A: RNDZ RDR - OPEN

V44, RR - SLEW

RT/ERR MON - LDG RDR/CMPTR

TM - H/H

22012

111:05

SS

111

+07

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	R	Ṙ
MAX Δ	2.27 NM	7 FPS
N78	.52	.6
VHF	.50	1.0
TM	32.00	

MSFN UPDATE

✓*COPY CSM CIRC P76

✓*COPY PADS FOR

✓* NO PDI + 12 ABORT

✓* PDI

✓* PDI EARLY ABORT

✓* PDI LATE ABORT

✓* T2 ABORT

✓* T3 TIG

✓* SHe PRESSURE

*SET DET TO COUNT DN TO DOI-2

IMU FINE ALIGN

V76

P52 OPT 3

CB(11) AC BUS B: AOT LAMP - CLOSE

AOT - DETENT F/0.0°

PGNS MODE CONT - AUTO

1ST STAR PROCYON (216)

PRO

2ND STAR ALDEBARAN (211)

N05 ANGLE DIFF 00003

PRO

N93 TORQUING ANG

X +00056

Y +00044

Z -00028

PRO, RCD GET ____:____:____

N25, (ENTR FOR NO COAS CAL)

COAS CALIBRATION

PRO, ENTR

N70, ENTR 011 (ALDBN), PRO

N87, (+00000, +00000) PRO, PRO

P00

DETENT CL

CB AOT LAMP - OPEN

BIAS AZ 0
EL 5.40

111:05

111:20

AGS CHECK
TO COAS CAL

DATE 11/7/72

PAGE 1-3

111:20

MSFN UPLINK TO DOI-2

22012 111:45

MSFN UPLINK, UPDATE

*UPDATA LINK - DATA *
 *UPLINK CSM/LM S.V., E-MEMORY *
 * DESCENT TARGETING *
 *UPDATA LINK - OFF *
 COPY AND LOAD GYRO DRIFT COMP, PIPA BIAS
 * (REF P2 DATA CARD BOOK) *

V47

*414+1 400+3 *

V83, SET ORDEAL *

*317R, 440R *

MNVR TO AGS CAL ATT

V49, +02250 OGA ROLL 24 }
 +06750 IGA PITCH 77 } FDAI
 +02250 MGA YAW 339 }

*VHF ANT - AFT *
 *VHF B XMTR - DATA *
 *VHF / WITH CSM *

AGS CALIBRATION

*READ AND RECORD INITIAL CAL. NOS. *
 PGNS MODE CONT - ATT HOLD *
 *VERIFY 25 MIN SINCE TURN-ON *
 V60, V76, V16N20E
 RATES < 0.075°/SEC
 *400+6 (647R > 00034) *
 *400R *

MONITOR ICDU LIMITS

{ OGA +00000 / +04500 }
 { IGA +04500 / +09000 } LIMITS
 { MGA +00000 / +04500 }

BEFORE LIMITS ARE EXCEEDED, 400+0.
 IF TIME IS LESS THAN 5 MINUTES
 REPEAT AGS CALIBRATION

*CHECK ECS, RCS, EPS, APS
 *CYCLE CWEA CB

111:45

*400R+0 *
 *READ AND RECORD CAL VALUES *

	INIT	CAL	Δ LIM	Δ NOM
540	-00002	- 3	± .039	0.0
541	+00002	+ 7	± .039	0.0
542	-00002	- 0	± .039	0.0
544	+00001	+ 32	±2.00	0.3
545	-00101	- 87	±2.00	0.3
546	-00023	- 63	±2.00	0.3

GO/NO GO FOR DOI-2

CONFIGURE COMM FOR LOS

*MATCH INDICATED ANGLES *
 *TRACK MODE - SLEW *
 *S-BD ANT - FWD *
 *SET P (+90) *
 * Y (-45) *
 *PCM - LO *
 *UPLINK SQUELCH - ENABLE *
 *COMM CHK WITH CSM *

DOI-2 PREP

P30 LOAD PAD
 SET DET, PRO
 P41
 N18 R,P,Y BYPASS MNVR
 N85
 MNVR TO N85 -VGX,0,0 (0,085,0)
 V77E
 N40E ✓DET, VG
 *410+5, 373+TIG *
 *450-VGX, 451+0, 452+VGZ *
 *310R ✓DET, 370R ✓VG *
 *500R *

CSM CIRC 111:52:3009 (111:55)

DOI-2 112:02:4092 (112:01) 5 MIN MAX SLIP

T=0 NULL VG'S TO .2FPS
 V82E ✓HP (7.2)

112:05

0 + .1 + .1 61.4 x 7.0 (6.7)

P76 (UPDATE CSM S.V.) TO MSFN UPLINK

112:05

P76 (UPDATE CSM S.V.) PRO
V82 ✓CSM HA/HP (70.3/54.3) (70.1 x 55.0)
*PCM - HI
V47
*414+1
V83
*317R, 440R
*PCM - LO
*373 _____ (+01696)(PDI)
*410+0

P63 IGNITION ALGORITHM TEST

P63
*RESET DET TO COUNT DN TO PDI
PGNS MODE CONT - AUTO
N18 R, P, Y (0, 108, XXX) PRO
YAW TO 290°
P00
V48, 22112, 00011, PRO, V34
*CAMERA SETTING (PDI)
*LM3/DAC/10/CEX-WDG (T/2.8, 1/500, ∞)
* 12 FPS, .75 MAG ~~60~~ (6 MIN)
*CAMERA SETTING (EARTH RISE)
*LM/DC/60/CEX (f16, 1/250, 74 ft)
* 5 FR, MAG A

COAS TO OVERHEAD WINDOW
VERIFY LOOSE GEAR STOWED
RESTRAINTS ATTACHED
VERIFY FDAI'S INERTIAL

PRE-PDI ECS CHECKOUT

HELMETS AND GLOVES ON
*CABIN REPRESS - CLOSE
*SUIT GAS DIVERter - EGRESS
*CABIN GAS RETURN - EGRESS
*PRESS REGS A&B - EGRESS

PRE-PDI SWITCH SETTING CHECK

*VHF ANT - FWD
CB(11)EPS: INV 1 - CLOSE
*SELECT INV 1

112:15

CB(11) STAB/CONT: AELD - CLOSE
CB(11) STAB/CONT: ABORT STAGE - CLOSE
RESET ENG STOP PB
SET WINDOW BARS

*CB(16) STAB/CONT: AELD - CLOSE
CB(16) STAB/CONT: ABORT STAGE - CLOSE
*CYCLE CWEA CB
-35 or Earlier *BATS 5 & 6 NORM FEED - ON
*RECORD GET 112 : 19 : 00
THROT CONT - AUTO
CDR TTCA - THROTTLE - MIN
*LMP TTCA - THROTTLE SOFT STOP
RATE SCALE - 25°/SEC
ATT/TRANSL - 4 JET
CHECK DPS, APS, RCS, ECS, EPS, RECORD APS DATA
CHECK SWITCH GUARDS
PRPLNT QTY MON - DES 1
✓DPS CONFIG CARD

✓ PDI RULES

AUDIO MODE (BOTH - VOX)
*S-BD ANT - FWD, VERIFY COMM
*S-BD P _____ (-33)
* Y _____ (+54)
*S-BD ANT - SLEW (>3.0)
*TRACK MODE - AUTO
*S-BD P _____ (-24)
* Y _____ (-3)
*VHF B XMTR - OFF
*BIOMED - LEFT, PCM - HI
*UPLINK SQUELCH - OFF
DOI-2 BURN REPORT
*CHECK ED BATTs AND REPORT
*VOICE ASC BATT ON TIME TO MSFN

MSFN UPLINK, UPDATE

*UPDATA LINK - DATA
*UPLINK LM S.V., RLS
*UPDATA LINK - VOICE BU
*COPY, LOAD AGS RLS (231) _____

AOS
112
+35

112:15

P76 TO MSFN
UPLINK

DATE 10/18/72

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AGS INITIALIZE TO PDI

22112

GO-AROUND

AGS INITIALIZE

V47

*414+1

V83

*317R, 440R

*240 + (231 RLS PAD)

*254+01944

*262-00143

*400+3

*400+1

MODE SEL - AGS

*X-PTR-LO MULT

POWERED DESCENT INITIATION

V25 N69E (IF NO UPLINK)

-8

P63

PGNS MODE CONT - AUTO (V77)

AGS MODE CONT - AUTO

*RESET DET

N18, R, P, Y (0, 108, 290)

VERIFY FDAI

-5

CB(11) PGNS: LDG RDR - CLOSE

✓ALT XMTR

-4

PRO - FINAL TRIM

ENTR, ✓DET

GO/NO-GO FOR PDI

COMM CHECK WITH CSM

RESET WATCH

-2:00

MASTER ARM - ON

MODE SEL - PGNS

*471R

-0:30

ENG ARM - DES

-0:07.5

ULLAGE

-0:05

PRO

0:00

PDI 112:49:235 (112:49:38)

+0:02

(NO IGN) - START PB - PUSH

+0:05

DES ENG CMD OVRD - ON

MASTER ARM - OFF

112 49 5187
+ 2.8

SECURE SYSTEMS

ENG ARM - OFF

MASTER ARM - OFF

P00

LR - OFF

ASC BATTS - OFF

PRLPNT QTY MON - OFF

AUDIO - PTT

ECS - CABIN MODE

HELMETS & GLOVES - OFF (OPTIONAL)

AGS - ATT HOLD

ALIGN IMU

P52 (SAME STARS)

400+3

MSFN UPDATE

COPY PADS FOR:

NO PDI + 12

PDI

PDI EARLY ABORT

PDI LATE ABORT

T2 ABORT

T3 TIG

AGS ABORT CONSTANTS

224,225,226,305,662,673

AGS T2 UPDATE - 254

AGS 373 UPDATE

MSFN UPLINK

DESCENT TARGETING, ABORT CONSTANTS,
V47 LM & CSM S.V.

CONFIGURE COMM FOR LOS

PICK UP WITH P63 IGNITION ALGORITHM TEST, P.5

Landed on the Moon
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Gene Cerny

-2:00 MASTER ARM-ON
MODE SEL-PGNS
471R

- :30 ENG ARM-DES
- :07.5 ULLAGE
- :05 PRO

+ :00 **PDI**

+ :02 (NO IGN) ---

START PB-PUSH
+ :05 DES ENG OVRD-ON
MASTER ARM-OFF

+ :26 THROTTLE UP
√T/W > 1.6
367R

+2 V21 N69E (DN RNG)

+4 ✓ ED BATTS
YAW TO 340°
V57E TO PERMIT
LR DATA
+5 V24 N69E (DN RNG,
X-RNG)

30K, YAW = 0

N68

EVAL MAN CONT

+8 V23 N69E (ALT)

223+00130 (E @ 13K)
360-0XXXXOE

TFI	θ	ΔHMAX	(-HMAX) -HDOT	H	DPS	SBD
0:00	108	1310	67.0	56500	95	-33/54
0:30	108		66.0	54500	95	
1:00	98	1410	69.0	52400	95	-23/55
1:30	93		69.0	50300	92	
2:00	89	1400	68.0	48300	87	-8/55
2:30	85		65.0	46300	82	
3:00	82	1310	63.0	44400	76	3/53
3:30	79		62.0	42500	71	
4:00	77	1230	63.0	40600	66	11/51
4:30	76		66.0	38800	61	
5:00	75	+19800	70.0	36200	55	-19/14
5:30	73	+19800	76.0	34000	50	
6:00	72	+19800	85.0	31100	45	-17/13
6:30	72	+19800	(516.0) 103.0	30800	39	
7:00	67	+19800	(480.0) 106.0	26900	34	-27/-3
7:30	63	+16000	(450.0) 107.0	23900	30	
8:00	57	+11200	(402.0) 133.0	19500	27	-20/-9
SEQ CAMERA - ON			(348.0)			
8:30	56	+8400	143.0	15000	24	
9:00	56	+8000	(288.0) 154.0	10700	21	-19/10

P64

P64 + 15 SEC:
NO THROTTLE DN
- ABORT

169+03400
0726

MODE CONT (PGNS)-ATT HOLD

P66

X-PNTR - LO MULT

BINGO FUEL
DES QTY LT+1+31

TOUCHDOWN

ENG STOP - PUSH
ENG ARM - OFF
PRO
DES ENG CMD OVRD - OFF
MODE CONT (PGNS)-ATT HLD
MODE CONT (AGS)-AUTO
413+1
RECYCLE PARKER VALVES
THROTTLE/JETS - JETS

H	(-HMAX) -HDOT	DPS	VH (362)
7000	(225.0) 169.0	18	258
6000	(206.0) 152.0	18	240
5000	(184.0) 132.0	17	223
4000	(161.0) 110.0	16	200
3000	(134.0) 88.0	15	173
2000	(103.0) 63.0	14	138
1000	(63.0) 35.0	12	85
500	(35.0) 17.0	11	43
400	(28.0) 14.0	11	31
300	(20.0) 9.0	10	16
200	(12.0) 5.0	9	-5

ABORT STAGE - PUSH
ENG ARM - ASC
ENG STOP - RESET
ENG START - PUSH
V22 N46 E,E
MODE CONT (BOTH) - AUTO

TD+3 THRU
T2 ABORT

TD +3 THRU T2 ABORT

N76 5515.7 V HORZ
19.5 V VERT
CROSS RNG (<8.1)
N74 TFI, YAW, PITCH

PAGE 1-8

*RECYCLE PARKER VALVES
PRPLNT TEMP PRESS MON - ASC, THEN DES
ASC He MON - CYCLE
*O2/H2O QTY MON - ASC 1, 2, THEN DES 2, 1 *
*SEQUENCE CAMERA - OFF *
CB(11) PGNS: LDG RDR - OPEN

223/360 TO +0 AS REQUIRED

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17:16 T1 STAY/NO STAY

NO STAY

V22 N46 E,E
MODE CONT (BOTH) - AUTO
ABORT STAGE - PUSH
ENG ARM - ASC
ENG STOP - RESET
ENG START - PUSH

STAY

*414+2
*400+4
*400+1 (DO NOT ENTR)

P68
ENG STOP - RESET
PRO
P12
MODE CONT (PGNS) - AUTO
N33 T-2 (113:14:23)

N43 20.21
30.75

22:45

NO STAY

T2 STAY/NO STAY AND GO/NO-GO FOR
DPS VENT

-2:00 ASC He SEL - BOTH
MASTER ARM - ON
ASC He PRESS - FIRE
ASC He REGS 1, 2 - OPEN
*A ASC FEED 2-OPEN (UNLESS CDR) *
*A MAIN SOV -CLOSE (BUSS LOSS) *
*B ASC FEED 2-OPEN (UNLESS LMP) *
*B MAIN SOV -CLOSE (BUSS LOSS) *
*CRSFD - CLOSED *
*BAT 1,3 - OFF *
*BAT 2,4 - OFF *
*CB(16) EPS:ASC ECA CONT-CLOSE *
*DES BAT - DEADFACE *
*SELECT ASC H2O TANK *
*DES O2 - CLOSE *
*ASC 1 O2 - OPEN *
*DES H2O - CLOSE *
*ASC H2O - OPEN *
*400+1, 367R *
- :10 ABORT STAGE - PUSH (AT T=0 FOR AGS)
ENG ARM - ASC
- :05 PRO
:00 *DET - RESET, RELEASE *
T2 : : (113:14:23)
+ :01 ENG START - PUSH

STAY

*TAPE RECORDER - OFF
AUDIO MODE - ICS/PTT
P00

DATE 11/3/72

FDAI AND OVERHEAD WINDOW ANGLES FOR MANUAL DESCENT ABORT

DPS/APS

1:00	260/0	4:30	0/LV	8:00	0/LV
2:24	SD(FDAI)	4:44	300/36	8:14	300/36
2:36	SD(OHW)	6:26	270/5	10:16	270/15
		7:32	250/0	14:30	250/0
		8:19	SD	14:43	SD
1:30	0/0	5:00	0/LV	8:30	0/LV
1:44	300/0	5:14	300/36	8:44	300/36
2:30	260/0	6:58	270/5	11:12	270/15
3:20	SD	8:20	250/0	15:18	250/0
		9:03	SD	15:31	SD
2:00	0/LV	5:30	0/LV	9:00	0/LV
2:14	300/36	5:44	300/36	9:14	300/36
3:00	300/0	7:34	270/10	12:10	270/16
3:18	260/0	9:04	250/0	16:04	250/0
4:16	SD	9:46	SD	16:18	SD
2:30	0/LV	6:00	0/LV	9:30	0/LV
2:44	300/36	6:14	300/36	9:44	300/36
3:46	300/0	8:04	270/11	12:56	270/16
4:02	260/0	9:50	250/0	16:44	250/0
5:12	SD	10:39	SD	16:58	SD
3:00	0/LV	6:30	0/LV	10:00	0/LV
3:14	300/36	6:44	300/36	10:14	300/36
4:34	270/0	8:34	270/11	13:16	270/16
5:10	250/0	11:04	250/0	17:32	SD
5:57	SD(FDAI)	11:47	SD		
6:06	SD(OHW)				
3:30	0/LV	7:00	0/LV	10:30	0/LV
3:44	300/36	7:14	300/36	10:44	300/36
5:24	270/0	9:06	270/14	13:48	270/16
5:54	250/0	12:26	250/0	18:05	SD
6:50	SD(FDAI)	12:53	SD		
7:02	SD(OHW)				
4:00	0/LV	7:30	0/LV	11:00	0/LV *
4:14	300/36	7:44	300/36	11:20	308/39
5:56	270/0	9:38	270/14	12:00	305/38
6:38	250/0	13:38	250/0	12:30	302/36
7:34	SD(FDAI)	13:53	SD		CONT MANUAL
7:40	SD(OHW)				ASCENT ANGLES

AT 2:00 MIN

ALL PITCH RATES
5°/SEC

* ESTABLISH POSITIVE
HDOT, THEN ABORT
STAGE

APS

1:00	260/0	4:30	0/LV	8:00	0/LV
2:18	SD/(FDAI)	5:10	300/36	8:40	300/36
2:36	SD(OHW)	6:22	270/10	12:06	270/14
		8:26	250/0	14:26	250/0
		8:51	SD	15:05	SD
1:30	0/LV	5:00	0/LV	8:30	0/LV
2:10	260/0	5:40	300/36	9:10	300/36
3:32	SD(FDAI)	7:08	270/10	13:00	270/14
3:52	SD(OHW)	9:20	250/0	15:02	250/0
		9:46	SD	15:50	SD
2:00	0/LV	5:30	0/LV	9:00	0/LV
2:40	300/0	6:10	300/36	9:40	300/36
3:10	260/0	7:54	270/12	13:54	270/14
4:24	SD(FDAI)	10:14	250/0	15:34	250/0
4:34	SD(OHW)	10:42	SD	16:31	SD
2:30	0/LV	6:00	0/LV	9:30	0/LV
3:10	300/0	6:40	300/36	10:10	300/36
4:00	250/0	8:44	270/14	14:30	270/14
5:18	SD	11:08	250/0	16:12	250/0
		11:37	SD	17:07	SD
3:00	0/LV	6:30	0/LV	10:00	0/LV
3:40	300/36	7:10	300/36	10:40	300/36
4:22	270/0	9:34	270/14	14:30	270/14
5:28	250/0	12:02	250/0	17:00	250/0
6:09	SD	12:33	SD	17:37	SD
3:30	0/LV	7:00	0/LV	10:30	0/LV
4:10	300/36	7:40	300/36	11:10	300/36
5:00	270/5	10:24	270/14	14:48	270/14
6:30	250/0	12:56	250/0	17:50	250/0
7:02	SD	13:28	SD	18:09	SD
4:00	0/LV	7:30	0/LV	11:00	0/LV *
4:40	300/36	8:10	300/36	11:20	308/39
5:40	270/5	11:12	270/14	12:00	305/38
7:28	250/0	13:48	250/0	12:30	302/36
7:56	SD	14:19	SD		CONT MANUAL
					ASCENT ANGLES

AT 2:00 MIN

MANUAL ABORT

ASCENT MONITOR

188:01:36

ASCENT

TIG-2 MASTER ARM - ON
AUDIO MODE (BOTH) - VOX
400+1E GUID STEERING
RESET WATCH
367R (604+0 FOR AGS)
START CAMERA
- :10 ABORT STAGE - PUSH (AT T=0
ENG ARM - ASC FOR AGS)
- :05 PRO
+ :00 **ASC** (188:03:15)
+ :01 ENG START-PUSH (IF AUTO IGN)
AGS MODE CONT - AUTO
CHECK S-BD ANT
+1:00 YAW RT 30°, 623+1
N76E (VTGT, HDOT, XRG)
V16 N77E (Tgo, VGY, VI)
+4:00 STOP CAMERA
500R
500 FPS MAIN SOV (2) - OPEN+
ASC FEED 2 (2) - CLOSE+
200 FPS ENG ARM-OFF(If IGN WAS AUTO)
0 FPS ABORT STAGE-RESET
ENG STOP - PUSH
410+5
PRO, NULL X < 2 FPS
PRO
STOP DET, RESET WATCH
COPY GET
ENG STOP - RESET
POO
MCC FOR TWEAK

FOR NO VOICE (TRIM <2 FPS)
PGNS,AGS DIFFER <10 FPS,
TRIM ACTIVE SYSTEM
PGNS,AGS DIFFER >10 FPS,
TRIM SYSTEM THAT AGREES
WITH RR
(10° IN OHW) (0° YAW)

V82

TFI	θ	OHW (0°YAW)	VGX	H DOT	H	SBD
0:00			1070.0	0.0	0	
0:10			890.0	52.0	300	
0:30	308	40	4820.0	88.0	1600	
1:00	305	38	4650.0	123.0	5000	141/-52
1:30	302	36	4470.0	149.0	9100	
2:00	299	34	4250.0	168.0	13900	145/-47
2:30	296	32	4000.0	181.0	19200	
3:00	293	30	3720.0	188.0	24700	148/-42
3:30	289	27	3400.0	188.0	30400	
4:00	286	25	3060.0	183.0	36000	152/-37
4:30	282	22	2680.0	173.0	41300	
5:00	278	20	2270.0	157.0	46300	156/-31
5:30	274	17	1840.0	136.0	50700	
6:00	269	14	1370.0	111.0	54400	160/-24
6:30	265	11	870.0	83.0	57300	
7:00	260	8	330.0	52.0	59300	164/-20
7:18	257	6	0.0	32.0	60100	166/-15

NO AUTO IGNITION
WITHIN 30 SEC:
1. GUID CONT - AGS
2. ACA HOT FIRE
3. AGS MODE CONT - AUTO
STILL NO IGNITION:
1. ABORT STAGE - RESET
2. ENG ARM - OFF

MANUAL ASCENT (WILL NOMINALLY
BE TARGETED 11.5 MIN LATE)
CONFIGURATION NOMINAL EXCEPT:
MODE CONT - ATT HOLD
PROFILE NOMINAL EXCEPT:
7-STEP FOR DIRECT MODE

TFI	FDAI	OHW
0:00	0	
0:15	305	38
2:00	295	31
3:00	290	28
4:00	285	24
5:00	275	18
6:00	265	11
7:00	260	8

PITCH RATES
5°/SEC

MSFN WILL CALL PITCH AND
ROLL BIAS COMMANDS FROM
GROUND TRACKING.

6+30 - MAIN SOV(2) - OPEN+
ASC FEED 2 (2) - CLOSE+

SHUTDOWN
ENGINE ARM OFF
STANDBY TO RESET ABORT STAGE
Pb AND DEPRESS ENGINE STOP
Pb ON CALL FROM MSFN

Landed on the Moon
aboard the Apollo 17 LM "Challenger"

[Signature]

MISSION APOLLO 17, OCTOBER 25, 1972

TIME	RANGE	RDOT
L0+5	148	1667
L0+6	160	825
L0+7	164	-157
INS	163	-471
1+00	158	-468
2+00	154	-464
3+00	149	-460
4+00	144	-455
5+00	140	-450
6+00	135	-444
7+00	131	-437
8+00	127	-431
9+00	123	-423
10+00	118	-416

INSERTION 188:10:32

V82

V76

AGS MODE CONT-ATT HOLD

RR MODE-LGC

RATE/ERR MON(2)-RNDZ RDR

SHFT/TRUN +5

RATE SCALE 5°/SEC

RNG/ALT MON-RNG/RNG RT

CB(11) PGNS RR-CLOSED

*VHF ANT-FWD

*400+2 Z-AXIS STEER

*410+4 TPI EXEC

*373+0537.5 TIG TPI

*616+00005 ULLAGE

*623+0

*COPY AGS DATA

AUDIO MODE(2)-ICS/PTT

/INV 2, CB INV 1-OPEN

CB(11) & (16) ED: LOGIC PWR-OPEN

CB(11) ECS CABIN FANT-CLOSE

+1 GO/NO-GO FOR TWEAK

P47 FDAI (0,257,30)

*404+0, 405+0, 406+0

*MONITOR 470, 471, 472

+3 TWEAK 188:13:32

ΔV'S

-4.0 -9.9 +1.0

12102

INSERTION THRU TPI

P47 FDAI (0,242,0) OR 10° OHW

*404+0, 405+0, 406+0 *

*MONITOR 470, 471, 472 *

42 LM BAILOUT @ L.O.+12:10

TIG 188:15:32

ΔVX 39.4

*EXT LTG-TRACK *

P20, AUTO MNVR

V80, MAX N49(2.00,12.0)

P34 TGT TPI

*VERIFY PGNS WITH MSFN *

*V47, 414+1, 400+3 *

*400+2 Z-AXIS STEER *

*417+1 (7417+0) *

*411+1 START AUTO(19,18) *

*310R SET DET *

*303R θ TPI *

V82

V83 SET ORDEAL (35NM)

*317R, 440R, 277R *

V48, 12012

LM WT

35 CSM BAILOUT GET P76 PAD

30 CHART R/RDOT

SS

27

M=15, V32

24

*COMPARE CMC, AGS, VHF/RR *

*POLAR PLOT @ 90 NM *

21

18

*CHECK RCS, EPS, ECS *

15

*514+0 *

*515+4 YAW STEER VEC *

*516+0 *

12

10 CHART R/RDOT/θ

9

8 PRO-FINAL COMP

*411+0 STOP AUTO *

6 *COMPARE CMC, AGS *

CHECK TIG OF CSM

*/DET & APS BURN CARD *

P42 N86

PERFORM YAW/ROLL MANEUVER

*MATCH INDICATED ANGLES *

*TRACK MODE-SLEW *

*S-BD ANT-AFT *

SET P (+132) *

Y (-2) *

*BIOMED-OFF, PCM-HI *

*UPLINK SQUELCH-ENABLE *

*404+0, 405+0, 406+0 *

*623+1 *

*400+1 GUID STEER

5

*410+5

*500R

1:00 AGS MODE CONT-AUTO

:30 ABORT STAGE PB-PUSH

:10 MANUAL ULLAGE

:05 PRO

:00 TPI 188:57:32

ABORT STAGE PB-RESET

NO IGNITION

ENG ARM-ASC

MANUAL START

MANUAL STOP 3 SEC

ENG ARM-OFF

NULL RESIDUALS

INSERTION
THRU TPI

TPI THRU DOCKING

MISSION APOLLO 17, SEPTEMBER 1, 1972
MAN AGS UPDATES

0 TPI 188:57:32
V76, AGS MODE CONT-ATT HOLD
P35 TGT MCC 1 ATT CONT-PULSE
MAX N49(0.80,5.0) MODE CONT-AUTO
V67 (+02000,+00020,+00005)
*400+0 *
*623+0 *
*417+1 (/621+0) *
*411+1 START AUTO(13,12) *
2 *410+4 TPI EXEC *
*373+TPI TIME +15 MIN *
*307+028.00 *
4 RDOT \$R
6 RDOT \$R
8 RDOT \$R
9 CHART 0 *
10 RDOT \$R
12 PRO FINAL COMP RDOT \$R
13 CHART R/RDOT/0
*411+0 STOP AUTO *
370R TOTAL VEL MCC1
371R ΔV TPF
*404+0, 405+0, 406+0 *
P41, V77
14 *410+5 ATT CONT-
*502R MODE CONT
05 *472R/502R *A/H
15 MCC1
NULL RESIDUALS

V76
P35 TGT MCC 2 ATT CONT-PULSE
V93 MODE CONT-AUTO
*VERIFY PGNS (PCM-HI) *
*V47, 414+1, 400+3 *
*411+1 START AUTO *
*EXT LTG-OFF *
SR
17 *410+4 TPI EXEC *
*373+TPI TIME +30 MIN *
*307+013.00 *
19 RDOT \$R
21 RDOT \$R
23 RDOT \$R
24 CHART 0 *
25 RDOT \$R
27 PRO-FINAL COMP RDOT \$R
28 CHART R/RDOT/0
*411+0 STOP AUTO *
370R TOTAL VEL MCC2
371R ΔV TPF
*404+0, 405+0, 406+0 *
P41, V77
29 *410+5 ATT CONT-
*502R MODE CONT
05 *472R/502R *A/H
30 MCC2
NULL RESIDUALS

P00
V48, 11002 11002
P47, V63
*404+0, 405+0, 406+0 *
S-BD ANT-AFT, VERIFY COMM
*S-BD P (+132) *
Y (-2) *
*S-BD ANT-SLEW (>3.0) *
*TRACK MODE-AUTO *
*BIOMED-LEFT, PCM-HI *
*UPLINK SQUELCH-OFF *
TPI BURN REPORT
40 INITIATE BRAKING
30 FPS - 6000 FT DES ATTACHED
20 FPS - 3000 FT 5 FPS - 1000 FT
10 FPS - 1500 FT 3 FPS - 500 FT
5 FPS - 600 FT 1 FPS - 100 FT
*SETUP CAMERA FOR *
* DOCKING: *
*LM3/DAC/10/CEX-ULC *
* (T8,1/250,6) 1FPS *
* .25 MAG(0), (4MIN) *
V34, P00
V76 ATT CONT-PULSE
MANEUVER/PICTURES OF SIMBAY,RCS
55 INITIATE DOCKING
COAS TO OVHD WINDOW
*EXT LTG-DOCK *
SHFT/TRUN ±50 *
V41N72 (+000,+320)
CB RR(2)-OPEN, V44
FDAI (180,282,300)
V77 ATT CONT-
MODE CONT
65 CONTACT MODE CONT
CONFIRM CAPTURE FROM CSM
Landed on the Moon
aboard the Apollo 17 LM "Challenger"

190:05

CONFIGURE S-BAND

- 1 Verify FWD Dump VLV - AUTO
CB(11): ECS CABIN FAN 1 - OPEN
TAPE RECORDER - OFF
CSM Maneuvers To JETT ATT, Proceed With
"PREP FOR TRANSFER" Until Maneuver Completed
- 2 Verify: Jettison Attitude (049,XXX/139,075)
CSM in ~~Narrow Deadband~~ Attitude Hold
- 3 S-BAND: PM,SEC,PRIM,VOICE,PCM,RANGE,RIGHT,HI
S-BD ANT - AFT, Verify COMM
TRACK MODE - SLEW
S-BD P _____ (+203)
Y _____ (+ 61)
S-BD ANT - SLEW "(Peak Until >3.9)"
(DO NOT PLACE TRACK MODE - AUTO)

190:16

PREP FOR TRANSFER

- 1 Window Shades (3) - Close
Install Crash Bars
Verify Tunnel Pressurized From CSM
OVHD Dump VLV - OPEN
- 2 Doff Helmets and Gloves
- 3 Install Purse (ISA Bottom Pkt)
Remove & Stow In Purse:
CWG Adapter W/Cap (2, Fwd LHSSC)
Lt/Wt Headsets (2, LHSSC)
Purge Vlv (1, Aft LHSSC)
Waist Tethers (2, Fwd RHSSC)
Neck Ring Dust Covers (2, AFT RHSSC)
Monocular (1, RHSSC)

- 4 Remove 16mm Mag From Camr & Place
In ISA Top pkt (bag)
Disconnect Lower ISA Hooks
Stow LHSSC Collection Bag Aft of Engine
Cover
- 5 When Tunnel/LM Pressures Equal,
OVHD DUMP VLV - AUTO
Verify PRESS REGS A&B - EGRESS
Place LEVA Bags On Floor, Right Side-Fwd
- 6 Open Hatch
Receive Probe From CMP, And Stow On Left
Hand Side Using Outboard (Double)
Restraint Cable
Receive Drogue From CMP and Stow Over
Probe Using Inboard (Single) Restraint
Cables Through Drogue Handles
- 7 Receive Vacuum Cleaner Assembly From CMP
Transfer LEVA Bags (2), Helmets, & IV Gloves
w/Liner to CMP
- 8 Vacuum PGA's
- 9 CB(11) AC BUS A: TAPE RCDR - Open
COMM: CDR AUDIO - Open
CB(16) COMM: DISP - Open
: S.E. AUDIO - Open
Disconnect DSEA & Place In Purse
CB(11) COMM: CDR AUDIO - Close
CB(16) COMM: DISP - Close
: S.E. AUDIO - Close

Landed on the Moon
aboard the Apollo 17 LM "Challenger"

Gene Cerny

DATE 11/27/72 (P/I)

PAGE 3-1

AOS

191

+36

- 10 Receive Decontamination Bags & Jett Bag
From CSM
Transfer Purse To CSM

- 11 Unstow, Vacuum/Wet Wipe as required and
Transfer to CSM:
(*Decontamination Bag Provided)
✓ 70MM Magazines (4, 3 & 2 in Bags, RHSSC)
(3 and 3 in Bags, Bot
Boot Compt)
✓ APK (1, LHMS - AFT SRC's)
Flag Kit (LHMS - AFT SRC's)
✓ OPS (W/Highest Source Pressure), Perform
Checkout per Decal
✓ *SEP DSEA (Upper Boot Compt)
✓ Neutron Flux (+Z27)
✓ Core Stems (+Z27)
✓ Flight Data File (Place In Jett Bag
and Transfer to CMP)

- ✓ *Collection Bags (1, LHSSC) (1, RHSSC) - A7
✓ *Collection Bag (1, Recharge Station) - PGA Bag
✓ BSLSS Rock Bag (+Z27) - A7
✓ ISA - A2
Collection Bag (Box AFT Eng Cover) - A9

- ✓ 12 Unstow SRC's (2), Vacuum and Transfer to CSM
Receive B5 & B6 from CSM and Stow in SRC Rack
- ✓ 13 Receive Purse (empty) from CSM
- ✓ 14 Receive CSM Jet Bag & stow Behind LMP Restraint
Cables
- 15 Stow used Fecal & Urine Bags in Purse
- 16 Transfer Vacuum Cleaner (Leave Bag in LM)
& Purse to CSM
- 17 Transfer unused food to CSM

T92:00MSFN UPLINK/UPDATE

- 1 UPDATA LINK - DATA
MSFN Uplinks LM State Vector, P30
EXT ΔV Load & P99 Erasable Loads (3)
- 2 Copy LM DAP Wt & DEORBIT Burn Pad
- 3 V48, 12021, PRO
N47 5185 LM WT (From MSFN)
PRO PAD
- 4 V47E, 414+1
- 5 400+3

T92:13TARGET PGNS

- 1 P30 Target Impact Burn
N45
PRO, P00

CONFIGURE AGS

- 1 404+0
405+0
406+0
470R
- 2 MCC-H GO/NO-GO For LM Closeout
- 3 Verify All Items in LM TO CM TRANSFER
LIST (Pg 3-6) Have been Transferred or
Will Be Transferred "ON CREW."

195 N33
38
1300
X -02246 N81
Y +00569
Z +01677
ΔV 02860
BT 158
FPA 048
138
075

SS

192

+25

LOS

192

+50

192:53

CONFIGURE LM FOR JETTISON

- 1 VERIFY CSM MIN DB/ATT HOLD
GUID CONT - PGNS
PGNS MODE CONT - AUTO (NO DAP Lt - OFF)
AGS MODE CONT - ATT HOLD
ATT CONT (3) - MODE CONT
Verify INV-2
- 2 VHF A: XMTR - VOICE/RANGE
: RCVR - OFF
VHF B: XMTR - OFF
: RCVR - ON
BIOMED - OFF
- 3 ASC FEED (4) - tb-bp
SYS A&B QUADS (8) - ENABLE; tb-gray
CRSFD - tb-bp
SYS A&B MAIN SOV (2) - tb-gray
- 4 SUIT CIRCUIT RELIEF - AUTO
SUIT ISOL VLV (Both) - SUIT DISC
CB(11) COMM: CDR AUDIO - Open
CB(16) COMM: S.E. AUDIO - Open
ECS: LCG PUMP - Open
Both Disconnect LM Hoses & Stow *SUIT FAN Δ P - Open*
- 5 S-BAND VOICE - OFF
Verify UPDATA LINK - DATA
- 6 Configure CB's Per Chart

Landed on the Moon
aboard the Apollo 17 LM "Challenger"

Gene Cerny

ACA REMOVAL

PERFORM ON BACKSIDE OF MOON

1. CB(11) STAB/CONT: ATT DIR CONT - Open
: ATCA (PGNS) - Open ✓
COMM: CDR AUDIO - Open
CB(16) COMM: S.E. AUDIO - Open ✓
STAB/CONT: ATCA (AGS) - Open
ACA PROP (BOTH) - DISABLE
ATTITUDE CONTROL SW's (3) - DIRECT ✓
2. CDR's ACA
Pull slack in cables
Unlock Thumblock (rotate aft)
Rotate ACA approx. 30 Deg. forward
Remove ACA bracket screws (2) - cross slotted
Lift ACA out and remove the cable clamp screws (2) - single slot.
Disconnect PJ709A and PJ708A
Install ACA Shorting Plug on PJ709A
Install ACA Dust Cap on PJ708A
3. LMP's ACA
Remove the ACA bracket screws (2) - single slot
Lift ACA out and remove cable clamp screws (2) - single slot.
Disconnect PJ709B and PJ708B
Cover PJ709B and PJ708B with tape
4. RECONFIGURE
ATTITUDE CONTROL SW's (3) - MODE CONTROL
CB(11) STAB CONT: ATCA (PGNS) - Close
CB(16) STAB CONT: ATCA (AGS) - Close
COMM: S.E. AUDIO - Close

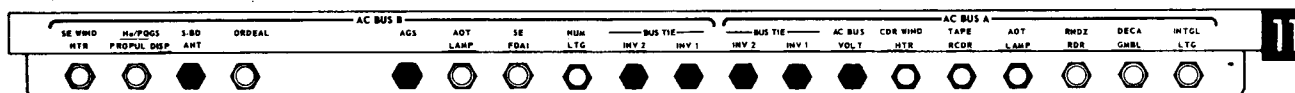
CONFIGURE FOR
LM JETT

DATE 9/27/72 11/17/72 (P1)

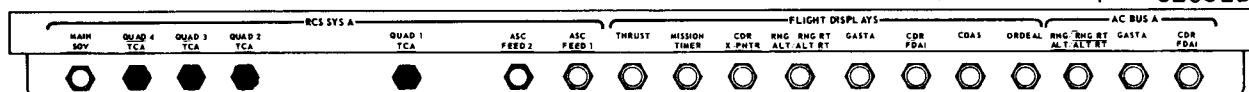
PAGE 3-3

POST DOCKING

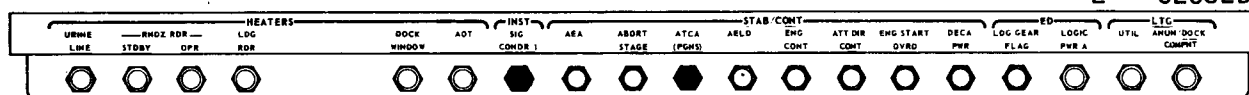
7 - CLOSED



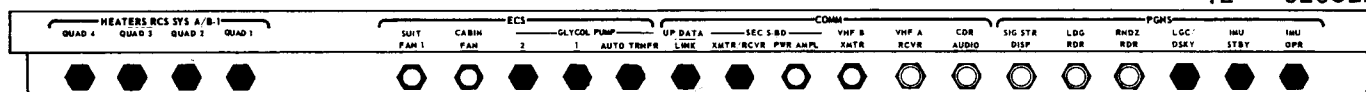
4 - CLOSED



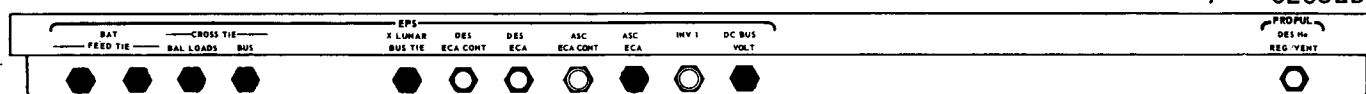
2 - CLOSED



12 - CLOSED



7 - CLOSED

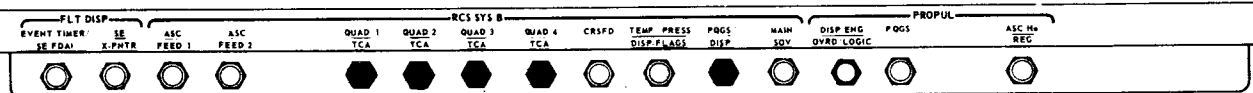


DATE 9/27/12 44777 (P.I.)
12/1/12

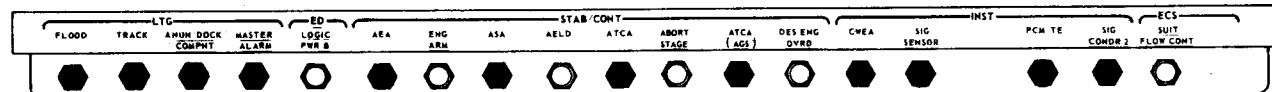
POST DOCKING

5 - CLOSED

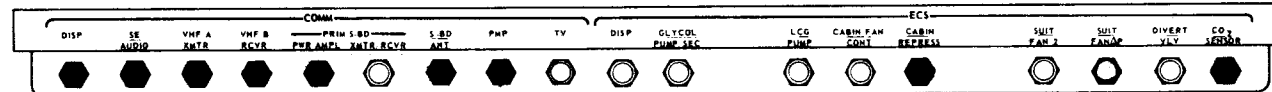
16



12
18 - CLOSED

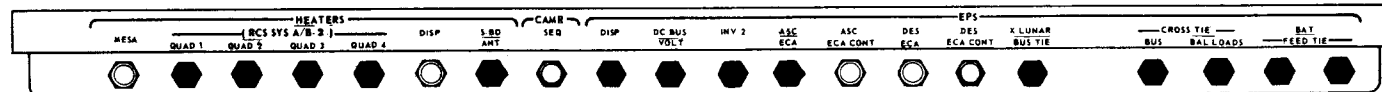


9 18 - CLOSED



OPEN ↗

14 - CLOSED





193:10

SR IVT TO CSM

193

+10

1 EXTERIOR LTG - TRACK
 BAT 5&6 BACK UP FEED-ON; tb(2) - Gray, *cb(16) LTG:*
 FLOOD LT - OFF
 Verify OVHD Dump VLV-AUTO
 Transfer To CSM

ANUN/DOCK/
COMPT-OPEN
MASTER ALARM - OPEN

LM TO CM TRANSFER LISTFloor LM

ITEM
 LEVA Bag (2) & Contents:
 LEVA
 EV Gloves w/Wrist mirrors
 Helmets (2)
 IV Gloves w/Liner (2 Pr)
 Purse & Contents (1st XFR):
 CWG Adptr w/cap (2)
 Lt/Wt Headsets (2)
 EMU Maint Kit (2)
 Purge vlv (1)
 Waist Tethers (2)
 LCG Plugs (2)
 PGA Pocket Contents
 PGA Elect Conn Covers
 Neck Ring Dust Covers
 Monocular (1)

Temp CSM
Stwg

HSB
 HSB
 TSB's
 A8
 A8
 L/H TSB
 L/H TSB
 L/H TSB
 L/H TSB
 PGA Bag
 PGA Bag
 U4

Temp A7
 R13
 R13
 A8
 R13
 A8
 Temp A7
 Temp A7

AFT Eng
 SRC Rack
 Temp Stwg

OPS
 SEP DSEA
 Neutron Flux
 Core Stems
 Flt Data File in Jett Bag
 Collection Bag (2)
 Collection Bag (1)
 BSLSS/ROCK Bag
 ISA & Contents:
 Lens Brush (1)
 16mm Mag(4) in Bag (EE)
 Cosmic Ray
 Collection Bags (2)
 Collection Bag
 SRC (2)
 Vacuum Cleaner
 Purse & Contents (2nd XFR):
 Used Urine Bags
 Used Fecal Bags
 Unused Food
 Suit (2) & Ancillary Eqpt:
 Bio Instrumentation (2)
 UCTA (2)
 FCS (2)
 LCG (2)
 Sunglasses in Pouch (2)
 Watch/Watchband (2)
 Pen (2)
 Pen-Felt Tip (2)
 Pencil (2)
 Pocket, C/L & Scissor (2)
 Pocket, Data (2)
 Penlight (2)
 Earplug (2) Pr
 Dosimeter - Personal (2)
 Passive (6)
 Comm Carrier (2)
 Scissor (1)

A7
 A7
 Wire Tray
 Wire Tray
 On A1
 A1
 PGA Bag
 A7
 A2

A9
 B5/B6
 RH TSB

Fecal Bag
 Fecal Bag
 B1

RHSSC
 RHSSC
 RHSSC
 Boot Compt
 Boot Compt
 LHMS
 LHMS

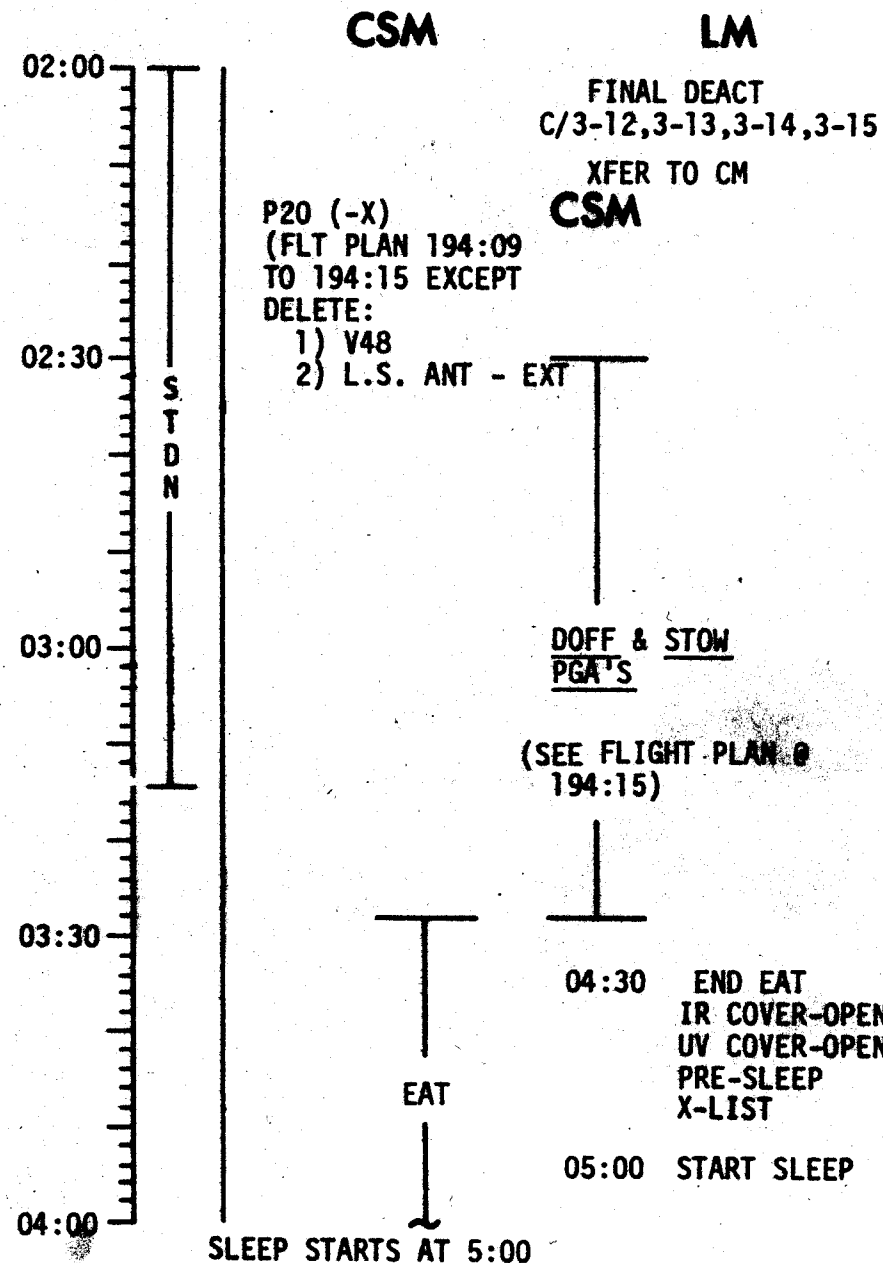
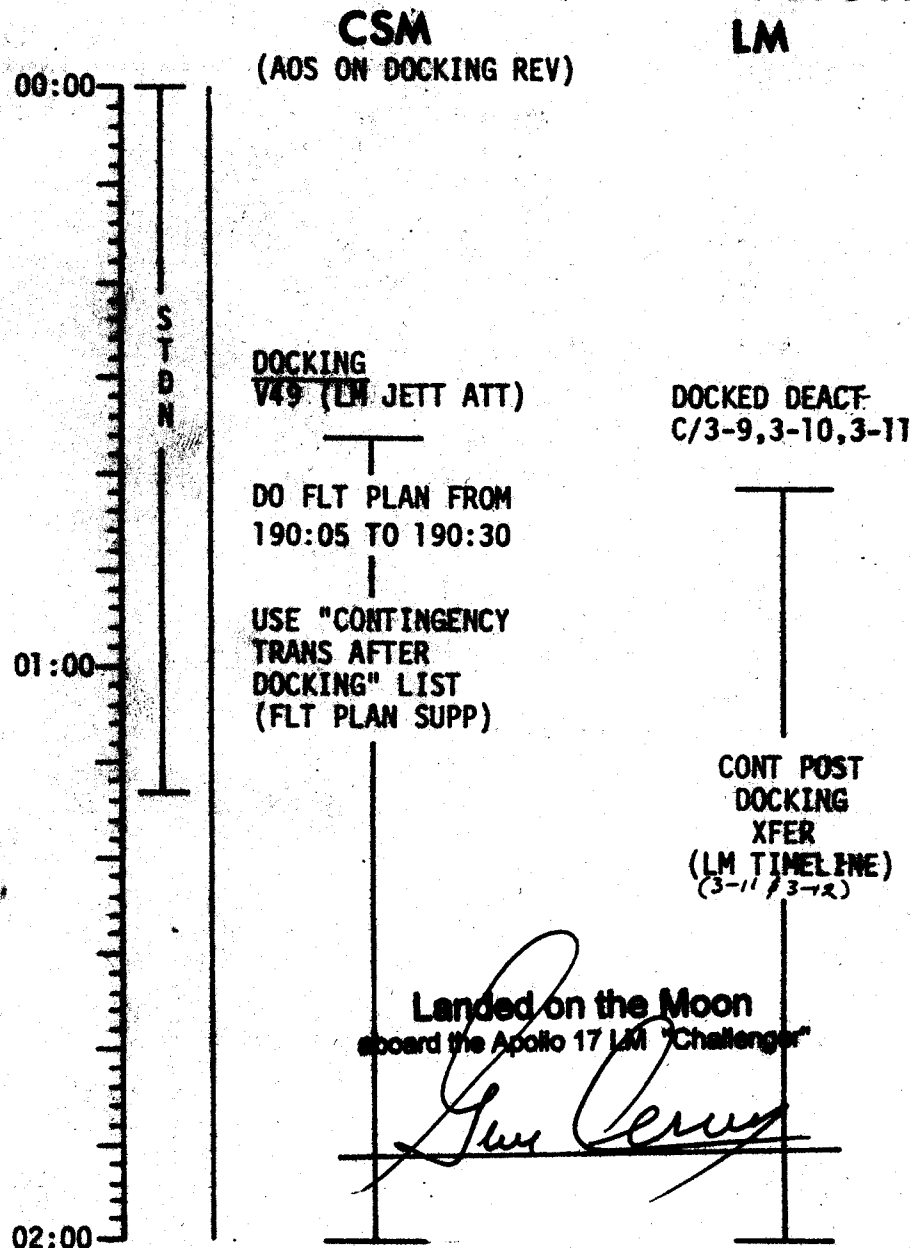
70mm Mag (4) in Bag
 70mm Mag (3) in Bag
 70mm Mag (2) in Bag
 70mm Mag (3) in Bag
 70mm Mag (3) in Bag
 APK (1)
 Flag Kit

Landed on the Moon

aboard the Apollo 17 LM "Challenger"

CONTINGENCY POST DOCKING XFER FLIGHT PLAN

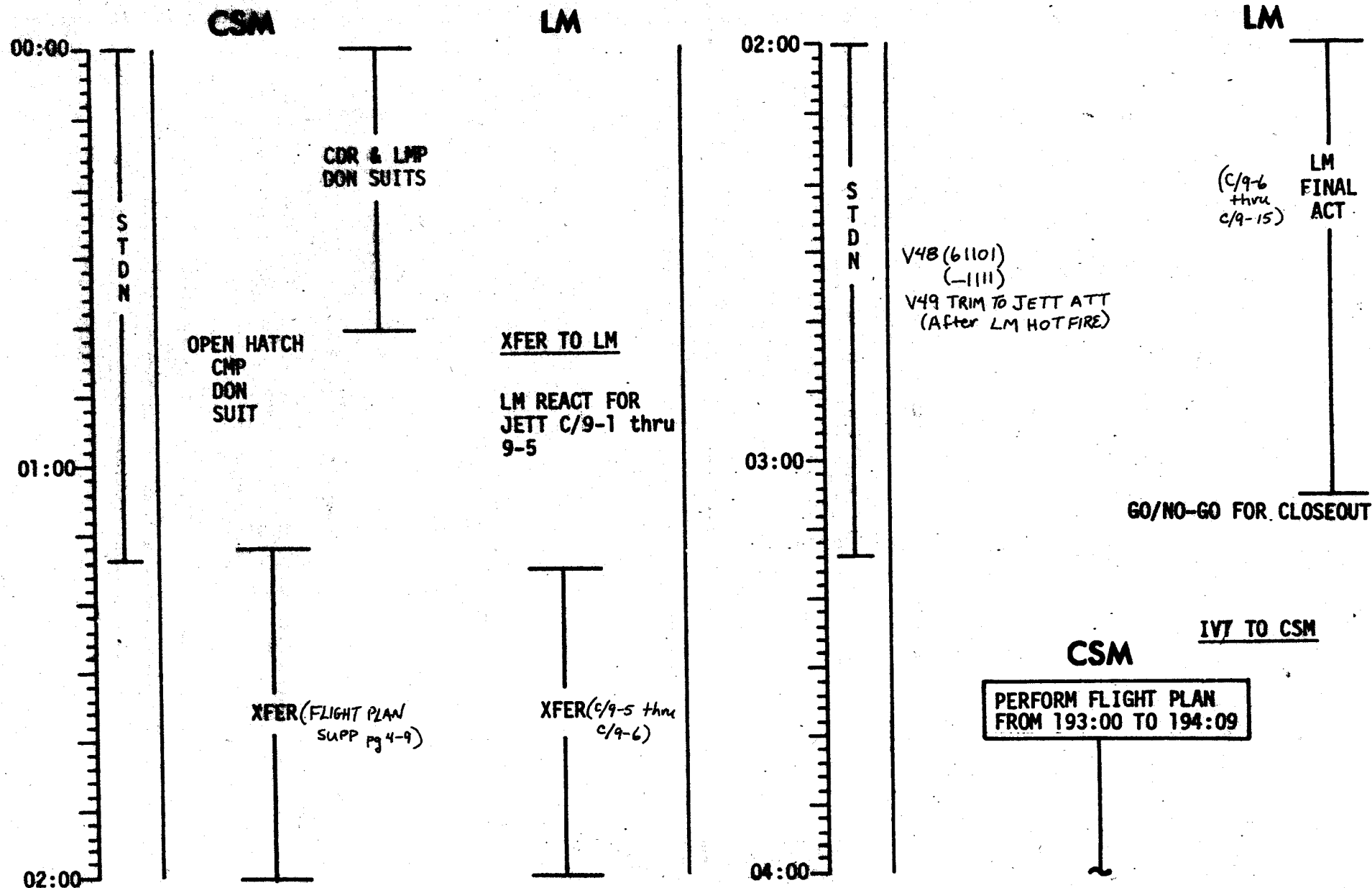
DATE 11/21/72 11/21/72 (P/E)



CONT POST
DOCKING TIMELINE

CONTINGENCY LM JETT
FLIGHT PLAN

PAGE 3-8



DATE ~~11/9/72~~ 11/27/72 (91)

CONTINGENCY LM JETT
FLIGHT PLAN

CSM

04:00

04:30

05:00

06:00

S
T
D
N

LM JETT
CSM SEP

INHIBIT ALL JETS EXCEPT:
A1 & C2 OR D1 & B2,A3,C4,B3,D4
V48 (11101)
(_1111)

P20 (-X SIM ATT)
N78 (+090.00)
(+052.25)
(+000.00)
N79 (+002.50)

AFTER IN ATTITUDE:
IR COVER - OPEN
UV COVER - OPEN

DOFF SUITS

Landed on the Moon
aboard the Apollo 17 LM "Challenger"

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CONTINGENCY POST DOCKING XFER

CONFIGURE S-BAND

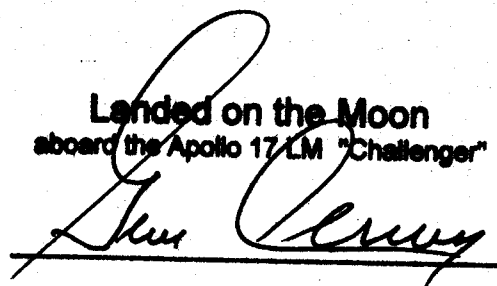
- 1 Verify FWD Dump VLV - AUTO
CB(11): ECS CABIN FAN 1 - OPEN
TAPE RECORDER - OFF
CSM Maneuvers To JETT ATT, ~~Proceed With~~
~~"PREP FOR TRANSFER" Until Maneuver Completed~~
MATCH S-BAND ~~X's~~
- 2 Verify: ~~Jettison Attitude (040, XXX/139, 075)~~
~~CSM in Narrow Deadband, Attitude Hold~~
TRACK MODE - SLEW
- 3 S-BAND: PM, SEC, PRIM, VOICE, PCM, RANGE, RIGHT, LO
S-BD ANT - AFT, Verify COMM
- 4 Perform DOCKED DEACTIVATION (STAGED)
LM CONTINGENCY C/L Pg 3-9, 3-10 & 3-11

PREP FOR TRANSFER

- 1 Verify Tunnel Pressurized From CSM
OVHD Dump VLV - OPEN
- 2 Doff Helmets and Gloves
- 3 Install Purse (ISA Bottom Pkt)
Remove & Stow In Purse:
CWG Adapter W/Cap (2, Fwd LHSSC)
Lt/Wt Headsets (2, LHSSC)
Purge Vlv (1, Aft LHSSC)
Waist Tethers (2, Fwd RHSSC)
Neck Ring Dust Covers (2, AFT RHSSC)
Monocular (1, RHSSC)

- 4 Remove 16mm Mag from Camr & place
In ISA Top pkt (bag)
Disconnect Lower ISA Hooks
Stow LHSSC Collection Bag Aft of Engine
Cover
- 5 When Tunnel/LM Pressures Equal,
OVHD DUMP VLV - AUTO
Verify PRESS REGS A&B - EGRESS
Place LEVA Bags On Floor, Right Side-Fwd
- 6 Open Hatch
Receive Probe From CMP, And Stow On Left
Hand Side Using Outboard (Double)
Restraint Cable
Receive Drogue From CMP and Stow Over
Probe Using Inboard (Single) Restraint
Cables Through Drogue Handles
- 7 Receive Vacuum Cleaner Assembly From CMP
Transfer LEVA Bags (2), Helmets, & IV Gloves
w/Liner to CMP
- 8 Vacuum PGA's
- 9 Receive Decontamination Bags
Transfer Purse to CSM

Landed on the Moon
aboard the Apollo 17 LM "Challenger"



- 10 Unstow, Vacuum/Wet Wipe as required and
Transfer to CSM:
(*Decontamination Bag Provided)
70MM Magazines (4, 3 & 2 in Bags, RHSSC)
 (3 and 3 in Bags, Bot
 Boot Compt)
*SEP DSEA (Upper Boot Compt)
Neutron Flux (+Z27)
Core Stems (+Z27)
*Collection Bags (1, LHSSC) (1, RHSSC) - A1
*Collection Bag (1, Recharge Station) - PGA Bag
*BSLSS Rock Bag (+Z27) - A7
*ISA - A2
Collection Bag (Box AFT Eng Cover) - A9

- 11 Unstow SRC's (2), Vacuum and Transfer to CSM
Receive B5 & B6 from CSM and Stow in SRC Rack
- 12 Receive CSM Jet Bag & stow behind LMP Restraint Cables
- 13 Transfer Vacuum Cleaner (Leave Bag in LM)

LM CLOSEOUT

- 1 MCC-H GO/NO-GO FOR LM CLOSEOUT
- 2 Perform FINAL DEACTIVATION Pg 3-12,3-13,3-14 & 3-15
(LM CONTINGENCY C/L)
- 14 Both Disconnect LM Hoses & Stow
Verify OVHD DUMP vlv - AUTO
TRANSFER TO CSM

RENDEZVOUS TIMELINES
RELATIVE MOTION TRAJECTORIES
INERTIAL PLOTS
AND
ABORT CHARTS

PDI SUMMARY DATA

8/30/72 Final

PAGE 4-2

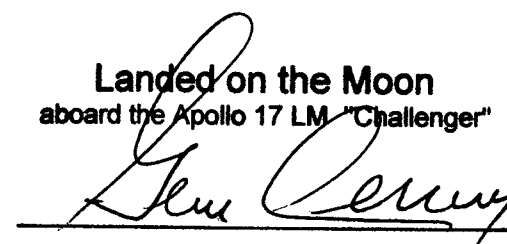
PAGE	ABORT	INS			BOOST	HAM	CSI		CDH			TPI	AIM		
		TIME PDI+	TIME PDI+	N76	TIME INS+	TIME INS+	TIME INS+	ΔVX	TIME INS+	ΔVX	ΔVZ	TIME PDI+	TIME PDI+	ΔVX	ΔVZ
A-1	NO 1+12	NA	NA	NA	NA	NA	1+07+00*	58.4	2+09+35*	-126.1	8.4	2+47+26	12+00	106.5	-50.0
	NO 1+12Δ	NA	NA	NA	NA	NA	1+07+00*	49.4	2+09+15*	-118.2	13.1	2+47+25	12+00	93.0	-50.0
A-2	1+00	2+05	5656.2	132.6/53784.	NA	NA	0+55+00	57.2	1+57+08	-115.2	-42.4	2+47+30	NA	NA	NA
	2+00	4+00	5651.9	131.8/58323.	↓	↓		55.5	1+57+03	-113.0	-38.2		↓	↓	↓
	3+00	5+43	5646.3	128.3/60018.				54.7	1+56+54	-108.5	-31.4				
	4+00	7+18	5639.5	122.9/60023.				54.4	1+56+39	-101.6	-22.5				
	5+00	8+49	5630.0	115.3/60030.				54.4	1+56+18	-92.1	-11.3				
	6+00	10+13	5617.5	105.5/60039.				54.5	1+55+51	-79.8	1.5				
	7+00	12+35	5596.3	91.9/64950.				53.1	1+55+14	-61.8	17.7				
A-3	8+00	14+27	5571.5	76.4/71046.	↓	↓	↓	51.5	1+54+31	-41.4	32.6	↓	↓	↓	↓
	9+00	16+06	5546.3	59.3/74249.				50.7	1+53+45	-18.2	45.2				
A-4	10+00	17+18	5564.0	71.7/72715.	50+00	1+50+00	2+40+00	38.9	3+39+14	-32.3	-59.4	4+46+26	NA	NA	NA
	11+00	18+21	5559.6	65.7/67727.	↓	↓	↓	41.8	3+39+01	-26.1	-43.6	↓	↓	↓	↓
	12+00	19+24	5555.9	59.9/62049.				44.5	3+38+49	-19.8	-29.3				
	13+00	20+27	5547.3	52.7/60251.				46.4	3+38+32	-11.5	-12.3				
	14+00	21+27	5539.8	47.2/60250.				47.2	3+38+19	-4.8	.1				
	15+00	22+26	5532.2	41.8/60248.				47.9	3+38+05	2.2	11.8				
	16+00	23+26	5524.6	36.3/60246.				48.2	3+37+51	9.3	22.1				
	17+00	24+25	5517.0	30.9/60244.				48.5	3+37+37	16.7	31.4				
A-5	T2-1	7+22Ω	5515.7	30.0/60154.	50+00	3+50+00	4+40+00	42.9	5+37+23	22.9	55.1	6+45+14	NA	NA	NA
A-6	NO 2+12	NA	NA	NA	1+12+00*	2+12+00*	3+12+00*	47.4	4+15+07*	-141.9	29.0	4+51+40	12+00	122.5	-50.0
	NO 2+12Δ	NA	NA	NA	1+12+00*	2+12+00*	3+12+00*	38.6	4+14+40*	-130.0	37.6	4+51+39	12+00	110.0	-50.0
A-7	1+00	2+07	5676.9	149.7/54128.	1+00+00	2+00+00	3+00+00	47.3	4+02+55	-136.4	-18.8	4+51+49	NA	NA	NA
	2+00	4+02	5672.6	149.1/59080.	↓	↓	↓	45.1	4+02+51	-134.8	-15.8	↓	↓	↓	↓
	3+00	5+44	5669.5	147.1/60019.				44.6	4+02+45	-131.8	-8.7				
	4+00	7+20	5666.3	144.5/60024.				44.2	4+02+37	-128.2	.5				
	5+00	8+50	5661.7	140.7/60031.	↓	↓	↓	43.8	4+02+26	-123.4	11.7	↓	↓	↓	↓
	6+00	10+15	5655.9	136.0/60040.				43.5	4+02+13	-117.2	25.2				
A-8	7+00	12+41	5665.0	146.4/65188.	NA	NA	0+55+00	50.4	1+57+38	-127.8	-57.1	2+52+54	NA	NA	NA
	8+00	14+33	5641.5	130.9/71212.				49.9	1+56+57	-109.7	-32.3				
	9+00	16+11	5617.9	113.9/74326.				50.0	1+56+12	-89.3	-8.6				
	10+00	17+20	5602.3	100.8/72751.				50.9	1+55+37	-72.9	7.4				
A-9	11+00	18+23	5590.1	88.7/67752.				52.3	1+55+05	-57.5	20.2				
	12+00	19+25	5578.4	76.7/62072.				53.8	1+54+33	-41.9	31.1				
	13+00	20+28	5560.6	62.5/60253.				54.3	1+53+54	-22.9	42.0				
A-10	14+00	21+27	5545.8	51.6/60250.	↓	↓	↓	54.2	1+53+25	-8.0	48.4	↓	↓	↓	↓
	15+00	22+26	5530.8	40.8/60247.				54.0	1+52+56	7.3	53.6				
A-4	T2-2	7+22Ω	5515.7	30.0/60154.	50+00	1+50+00	2+40+00	48.4	3+37+34	17.8	32.9	4+51+43	NA	NA	NA

Ω INDICATES TIME IS REFERENCED TO LIFT-OFF.

* INDICATES TIME IS REFERENCED TO PDI.

Δ ASSUMES NO DOI-2

Landed on the Moon
 aboard the Apollo 17 LM "Challenger"



DATE 9/8/72

RANGE AND RANGE RATE AT INS AND 10 MINUTES PRIOR TO SUBSEQUENT BURNS

8/30/72 Final

PAGE	ABORT TIME PDI+	INS		BOOST		HAM		CSI		CDH	
		RANGE	RANGE RATE	RANGE	RANGE RATE	RANGE	RANGE RATE	RANGE	RANGE RATE	RANGE	RANGE RATE
A-1	NO 1+12 NO 1+12 Δ	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA	177.3 151.6	-592.2 -536.6	97.7 97.2	-167.9 -172.5
A-2	01+00	133.5	570.0	NA	NA	NA	NA	163.7	-495.8	106.5	-125.8
	02+00	130.6	555.2					158.8	-485.4	101.8	-126.1
	03+00	117.5	526.6					144.1	-460.1	100.2	-131.8
	04+00	96.1	472.7					121.9	-417.4	99.9	-144.9
	05+00	69.4	331.8					91.7	-346.6	99.4	-138.8
	06+00	53.2	-86.6					55.6	-208.5	98.0	-150.2
	07+00	78.7	-454.3					30.8	203.1	94.1	-159.3
A-3	08+00	144.9	-497.4					74.9	215.5	92.3	-163.1
	09+00	219.9	-473.2					135.3	72.1	90.2	-170.1
A-4	10+00	283.7	-452.9	216.8	152.4	188.7	-390.4	37.3	47.2	97.5	-67.3
	11+00	339.9	-481.8	261.2	91.3	214.3	-405.6	69.7	28.9	98.6	-79.1
	12+00	397.3	-474.7	314.2	49.3	241.6	-420.7	96.0	-0.4	99.6	-93.8
	13+00	462.1	-459.9	374.5	-2.8	271.6	-430.2	126.2	-37.5	99.9	-105.7
	14+00	511.9	-445.3	420.8	-45.6	294.2	-433.3	149.3	-69.2	98.1	-116.7
	15+00	561.4	-430.6	466.5	-88.9	287.1	-426.7	172.1	-103.0	96.3	-126.2
	16+00	610.5	-415.7	511.7	-132.7	339.7	-432.6	194.5	-138.8	94.1	-133.9
	17+00	659.3	-400.7	556.5	-177.3	362.5	-429.2	216.3	-177.0	92.3	-142.2
A-5	T2-1	1024.9	-334.7	934.1	-187.4	373.0	-394.8	244.9	-200.9	88.3	-174.6
A-6	NO 2+12	NA	NA	398.8	-815.2	174.9	562.0	208.2	-693.5	99.9	-192.0
	NO 2+12 Δ	NA	NA	345.9	-742.6	149.1	500.3	169.1	-613.2	98.9	200.0
A-7	01+00	381.6	666.8	385.8	-738.2	143.5	516.3	209.7	-635.0	107.4	-160.7
	02+00	379.2	652.6	380.7	-728.7	144.8	506.5	206.4	-627.7	103.8	-146.4
	03+00	365.7	643.2	365.2	-717.7	138.3	504.6	196.5	-615.3	101.7	-166.5
	04+00	341.8	634.0	340.7	-699.7	127.6	505.6	182.6	-596.5	100.3	-174.0
	05+00	307.5	622.8	306.6	-673.9	112.1	506.1	163.9	-568.0	99.2	-183.9
	06+00	262.8	608.2	263.1	-638.0	92.9	500.8	141.2	-527.2	98.8	-184.6
A-8	07+00	208.1	599.4	NA	NA	NA	NA	224.3	-591.1	101.6	-125.6
	08+00	135.1	530.3					155.3	-480.1	97.9	-131.0
	09+00	68.8	337.1					88.6	-337.2	95.4	-138.2
	10+00	53.8	-246.9					41.5	-105.0	94.6	-144.2
A-9	11+00	92.5	-482.8					36.0	278.7	94.6	-157.2
	12+00	145.4	-512.8					72.8	227.6	95.3	-163.7
	13+00	209.0	-502.4					122.7	108.1	92.9	-170.5
A-10	14+00	259.0	-484.8					161.7	15.0	92.2	-174.9
	15+00	309.2	-465.0					200.3	-78.7	89.7	-175.3
A-4	T2-2	677.7	-396.9	573.6	-188.3	374.1	-428.6	227.5	-187.1	96.6	-149.3

 Δ ASSUMES NO DOT-2

INSERTION THRU BOOST

MISSION APOLLO 17, OCTOBER 6, 1972

60 INSERTION

SS

V82
AGS MODE CONT-ATT HOLD
SHFT/TRUN +5
RATE SCALE 5°/SEC
RNG/ALT MON-RNG/RNG RT
*VHF ANT-FWD
*EXT LTG-TRACK
*SEQUENCE CAMERA-OFF
*400+2
*616+00005 ULLAGE
*623+0
*RATE/ERR MON-RNDZ RDR
AUDIO MODE(2)-ICS/PTT
/INV 2, CB INV 1-OPEN
CB(11) & (16) ED: LOGIC PWR-OPEN
CB PGNS LDG RDR-OPEN
CB RR(2)-CLOSE

22112
ATT CONT-
PULSE
MODE CONT-
AUTO

+3 TWEAK

ΔVX

V48, 1 (2) 1002

1(2)1002

V41N72 (+000, +283)
CB RR(2)-OPEN, V44
RATE/ERR MON-LDG RDR/CMPTR
P52 OPT 3
CB AOT LAMP-CLOSE
AOT DETENT F/O°
V76
1st STAR
2nd STAR
N05 ANG DIFF
PRO
N93 TORQUING ANG
X
Y
Z
PRO N25(R1=14) GET

LOS

40 PRO N25(R1=15)
PRO TO PICAPAIR
DETENT CL
CB AOT LAMP-OPEN
V34
P00

*VERIFY PGNS WITH MSFN *
*V47, 414+1, 400+3 *
*400+2 *

V48, 1 (2) 2022

1(2)2022

*MATCH INDICATED ANGLES *
*TRACK MODE-SLEW *
*S-BD ANT-AFT *
SET P
Y
*BIOMED-OFF, PCM-HI *
*UPLINK SQUELCH-ENABLE *

30

Landed on the Moon
aboard the Apollo 17 LM "Challenger"

Gene Cerny

20

20

18

*CHECK RCS, EPS, ECS *

10

*VERIFY PGNS (PCM-HI) *
*V47, 414+1, 400+3 *
*400+2 *

SR

*EXT LTG-OFF *

ΔVX = +10.0 (HORZ)

P30

N33 TIG BOOST (INS + ΔT)

*373 + TIG BOOST *

IF STAGING @ BOOST

V48, 12022, PRO, V34

P41, V77

*404+0, 405+0, 406+0 *

*400+1 GUID STEER

*410+5 LOAD ΔV

*500R

ATT CONT-
MODE CONT

5

:05

*500R

STAGE AT BOOST IGNITION

0 BOOST

*A/H

BOOST THRU HAM

MISSION APOLLO 17, SEPTEMBER 1, 1972

60 BOOST

P00
V82
V76ATT CONT-
PULSE
MODE CONT-
AUTO

*400+2 Z-AXIS STEER *
 *416+1 1/2 PERIOD *
 *410+1 TGT CSI *
 *373+ TIG CSI *
 *275+ TIG TPI *
 *605+00777 COT *
 *310R SET DET (-50 MIN) *
 *402R ΔH *

50

AOS

S-BD ANT-FWD, VERIFY COMM
 */S-BD P *
 Y *
 *S-BD ANT-SLEW (>3.0) *
 *TRACK MODE-AUTO *
 *BIOMED-LEFT, PCM-HI *
 *UPLINK SQUELCH-OFF *

Landed on the Moon

about the Apollo 17 LM "Challenger"

40

40

V48, 12012
 (IF R < 400)
 CB RR(2)-CLOSE *
 RATE/ERR MON-RNDZ RDR
 RR MODE-LGC
 P20, AUTO MNVR
 V80, MAX N49(2.00,12.0)

12012

RR-AUTO
TRACK

P32, TGT CSI
 N11 TIG CSI (INS + ΔT)
 N37 TIG TPI (PDI + ΔT)
 [*VERIFY PGNS WITH MSFN *]
 [*V47, 414+1, 400+3 *]
 [*400+2 Z-AXIS STEER *]
 [*417+1 (7417+0) *]
 [*411+1 START AUTO(19,19) *]

V83 SET ORDEAL
 *317R, 440R, 277R *
 *COMPARE VHF/RR *

36

R

33

RDOT R

30

RDOT R

27

RDOT R

M=10, V32

24

RDOT R

21

RDOT R

18

*CHECK RCS, EPS, ECS *

RDOT R

15

V90

RDOT R

12

RDOT R

*402R *

10 PRO-FINAL COMP

*411+0 STOP AUTO
 *USE HAM CHART
 *COMPARE CMC HAM

V83, SET ORDEAL

*317R, 440R, 277R *

P30

N33 TIG HAM (INS + ΔT)

*373+ TIG HAM *

P41, V77, N86

*400+1 GUID STEER

*410+5 LOAD ΔV

*370R TOT ΔV

*500R

:05

*500R

A/H

O HAM

P00

V82

BOOST THRU HAM

INS/HAM THRU CSI

TIME	RANGE	RDOT
INS	288	-477
1+00	283	-473
2+00	278	-468
3+00	274	-462
4+00	269	-456
5+00	265	-449
6+00	260	-442
7+00	256	-434
8+00	252	-426
9+00	248	-418
10+00	243	-408

12102

NOM COELLIPTIC/ONE REV ABORTS

INSERTION

V82

AGS MODE CONT-ATT HOLD

RR MODE-LGC

SHFT/TRUN +5

RATE SCALE 5°/SEC

RNG/ALT MON-RNG/RNG RT

*VHF ANT-FWD

*SEQUENCE CAMERA-OFF

*616+00005 ULLAGE

*605+00777 COT

*RATE/ERR MON-RNDZ RDR

AUDIO MODE(2)-ICS/PTT

✓INV 2, CB INV 1-OPEN

CB(11) & (16) ED: LOGIC PWR-OPEN

CB(11) ECS CABIN FANT-CLOSE

CB RR(2)-CLOSE

+3 TWEAK

AVX

*VHF A: XMTR-VOICE/RNG

*400+2 Z-AXIS STEER

*507+0 Z-AXIS TRACT

*623+0

*410+1 TGT CSI

*373+ TIG CSI

*275+ TIG TPI

*416+1 1/2 PERIOD

*310R SET DET

*COPY AGS DATA(450R)

ATT CONT-
PULSE
MODE CONT-
AUTO

V48, 11002
V41N72 (+000, +283)
CB RR(2)-OPEN, V44
RATE/ERR MON-LDG RDR/CMPTR
P52 OPT 3
CB AOT LAMP-CLOSE
AOT DETENT F/0°
V76
1st STAR ALDEBARAN (11)
2nd STAR PROCYON (16)

RR-AUTO
TRACK417+1
R

RDOT R

RDOT R

45
42
39
N05 ANG DIFF
PRO
N93 TORQUING ANG

X

Y

Z

PRO N25(R1=14) GET

PRO N25(R1=15)

PRO TO PICAPAIR

DETENT CL

CB AOT LAMP-OPEN

V34

36 *EXT LTG-TRACK

RDOT R

V48, 12012

12012

CB RR(2)-CLOSE

RATE/ERR MON-RNDZ RDR

P20, AUTO MNVR

V80, MAX N49(2.00,12.0)

P32, TGT CSI

*VERIFY PGNS WITH MSFN

*V47, 414+1, 400+3

*400+2

*417+1 (✓417+0, 621+0)

45 *411+1 START AUTO(19,19)*

V83 SET ORDEAL (35NM)

*317R, 440R, 277R

RDOT R

33

MISSION APOLLO 17, OCTOBER 6, 1972

30 CHART RDOT RDOT R

27 RDOT R

24 RDOT R

M=10, V32

21 RDOT R

20 CHART RDOT

*COMPARE CMC, AGS, VHF/RR *

18 RDOT R

*CHECK RCS, EPS, ECS *

15 RDOT R

V90 OBTAIN CMC LM YDOT

12 RDOT R

10 CHART R/RDOT *

PRO-FINAL COMP

N81 LOAD CMC LM YDOT(IF>5fps)

9 RDOT R

N13 CDH TIG TO CSM

*MATCH INDICATED ANGLES *

*TRACK MODE-SLEW *

*S-BD ANT-AFT *

SET P (-16) *

Y (-28) *

*BIOMED-OFF, PCM-HI *

*UPLINK SQUELCH-ENABLE *

*411+0 STOP AUTO *

*COPY AGS DATA *

CB(11) ECS CABIN FANT-OPEN

V83 SET ORDEAL

*317R, 440R, 277R *

P41, V77, N86

5 *410+5 ATT CONT-
MODE CONT

*370R TOT ΔV

*ΔV's TO CSM *

:05 *500R/502R A/H

:00 CSI

NULL RESIDUALS

CSI THRU CDH

MISSION APOLLO 17, OCTOBER 6, 1972

58 CSI
 V76
 P33 TGT CDH
 MAX N49(0.80,5.0)
 V67 (+02000,+00020,+00005)
 ATT CONT-PULSE
 MODE CONT-AUTO

*417+1 (✓621+0) *
 *410+2 TGT CDH *
 *373R TM CDH (RECORD)
 *310R SET DET *
 *COPY AGS DATA *

55 *411+1 START AUTO(19,19)*

V82

V83 SET ORDEAL (45NM)
 *317R, 440R, 277R *

54 R

SR

51 M=7, V32 R

48 V90, LOAD CDH-30 R
 *COMPARE CMC,AGS,VHF/RR *

45 R

M=15, V32

42 R

39 V34, P30 (PC ONLY) R
 38 *411+0 STOP AUTO *

V90 LOAD CDH-30
 OBTAIN CMC LM YDOT

36 CHART RDOT R

P41, V77

*373+ TIG PC *
 35 *410+5 LOAD ΔV ATT CONT-
 *263R MODE CONT
 *501R *

:05 *270R/501R *A/H
 30 PLANE CHANGE

V76
 P33 TGT CDH ATT CONT-PULSE
 V93 MODE CONT-AUTO

*VERIFY PGNS WITH MSFN *
 *V47, 414+1, 400+3 *
 *400+2 Z-AXIS STEER *
 *410+2 TGT CDH *
 373+ TIG CDH IF PC
 *451+0 YDOT *
 *COPY AGS DATA *
 28 *411+1 START AUTO *

27 R

S-BD ANT-FWD,VERIFY COMM
 */S-BD P (-16) *
 Y (-28) *
 *S-BD ANT-SLEW (>3.0) *
 *TRACK MODE-AUTO *
 *BIOMED-LEFT, PCM-HI *
 *UPLINK SQUELCH-OFF *

CSI BURN REPORT
 TIG,ΔV'S, RESIDUALS

24 R

23 CHART RDOT

21 R

M=7, V32

*COMPARE CMC,AGS,VHF/RR *

18 R

*CHECK RCS, EPS, ECS *

15 R

V90 OBTAIN CMC LM YDOT

12 R

10 CHART RDOT *
 PRO-FINAL COMP

N81 LOAD CMC LM YDOT

9 R

*411+0 STOP AUTO

*VERIFY PGNS (PCM-HI) *

*V47, 414+1, 400+3 *

*400+2 Z-AXIS STEER *

*COPY AGS DATA *

V83, SET ORDEAL
 *317R, 440R, 277R *

P41, V77, N86
 (LARGE ΔV USE APS OR DPS)

5 *410+5 ATT CONT-
 *370R TOT ΔV MODE CONT
 *500R *
 *502R *

:05 *500R/502R *A/H

:00 CDH
 NULL RESIDUALS

53 CDH
 V76
 P34 TGT TPI ATT CONT-PULSE
 MAX N49(0.80,5.0) MODE CONT-AUTO
 V93
 *417+1 (/621+0) *
 *SET DET *
 V82
 *410+3 TPI SRCH *
 *310+TIME TO TPI *
 *303R θ TPI *
 *410+4 (When 303=26.6) *
 43 *411+1 START AUTO(19,19)*
 *POLAR PLOT @ 75 NM *
 310R
 42 3R
 39 M=7, V32 RDOT 3R
 *COMPARE TPI TIME *
 36 RDOT 3R
 33 RDOT 3R
 30 RDOT 3R
 SS
 27 RDOT 3R
 *MONITOR 303R θ TPI AND *
 *RETARGET IF REQ *
 *COPY AGS DATA *
 24 RDOT 3R
 M=15, V32
 *COMPARE CMC, AGS, VHF/RR *
 *VOICE LM TPI TIME *
 * TO CSM *
 21 RDOT 3R
 18 RDOT 3R
 *CHECK RCS, EPS, ECS *

15 RDOT 3R
 12 RDOT 3R
 *VERIFY PGNS (PCM-HI) *
 *411+0 STOP AUTO *
 *V47, 414+1, 400+3 *
 *400+2 Z-AXIS STEER *
 10 PRO-FINAL COMP
 TIG TO CSM
 *SET DET *
 9 CHART θ RDOT 3R
 *411+0 STOP AUTO *
 *410+3 *
 *310+TIME TO TPI *
 *303R θ TPI *
 *410+4 (WHEN 303=26.6) *
 *310R / WITH DET *
 *COPY AGS DATA *
 5 CHART R/RDOT/ θ
 *MATCH INDICATED ANGLES *
 *TRACK MODE-SLEW *
 *S-BD ANT-AFT *
 SET P *
 Y *
 *BIOMED-OFF, PCM-HI *
 *UPLINK SQUELCH-ENABLE *
 P41, V77, N86
 *404+0, 405+0, 406+0 *
 4 *410+5 LOAD ΔV & TIG *
 *507+1 THR Z-AXIS ATT CONT-
 *502R MODE CONT
 :05 *472R/502R *A/H
 :00 TPI
 NULL RESIDUALS

Landed on the Moon
 aboard the Apollo 17 LM "Challenger"

Gene Cerny

TPI THRU DOCKING

0 TPI
V76
*507+0 Z-AXIS TRACT *
P35 TGT MCC 1 ATT CONT-PULSE
MAX N49(0.80,5.0) MODE CONT-AUTO
V93

*417+1 (/621+0) *
*411+1 START AUTO(13,12) *

2 *410+4 TPI EXEC *
*373+TPI TIME +15 MIN *
*307+028.00 *

4 RDOT }R

6 RDOT }R

8 RDOT }R

9 CHART 0 *

10 RDOT }R

12 PRO FINAL COMP RDOT }R

13 CHART R/RDOT/0
*411+0 STOP AUTO *
370R TOTAL VEL MCC1
371R ΔV TPF

*404+0, 405+0, 406+0 *
P41, V77

14 *410+5 ATT CONT-
*502R MODE CONT

05 *472R/502R *A/H
15 MCC1
NULL RESIDUALS

V76
P35 TGT MCC 2 ATT CONT-PULSE
V93 MODE CONT-AUTO

*VERIFY PGNS (PCM-HI) *
*V47,414+1,400+3 *
*400+2 Z-AXIS STEER *
*411+1 START AUTO *

17 *EXT LTG-OFF *
*410+4 TPI EXEC *
*373+TPI TIME +30 MIN *
*307+013.00 *

19 RDOT }R

21 RDOT }R

23 RDOT }R

24 CHART 0 *

25 RDOT }R

27 PRO-FINAL COMP RDOT }R

28 CHART R/RDOT/0
*411+0 STOP AUTO *
370R TOTAL VEL MCC2
371R ΔV TPF

*404+0, 405+0, 406+0 *
P41, V77

29 *410+5 ATT CONT-
*502R MODE CONT

:05 *472R/502R *A/H
30 MCC2
NULL RESIDUALS

MISSION APOLLO 17, OCTOBER 6, 1972

CB(11) ECS CABIN FANI-CLOSE

P00

V48, 11002

11002

P47, V63

*404+0, 405+0, 406+0 *

S-BD ANT-FWD,VERIFY COMM

*S-BD P *

Y *

*S-BD ANT-SLEW (>3.0) *

*TRACK MODE-AUTO *

*BIOMED-LEFT, PCM-HI *

*UPLINK SQUELCH-OFF *

TPI BURN REPORT

40 INITIATE BRAKING

30 FPS - 6000 FT DES ATTACHED
20 FPS - 3000 FT 5 FPS - 1000 FT
10 FPS - 1500 FT 3 FPS - 500 FT
5 FPS - 600 FT 1 FPS - 100 FT

*SETUP CAMERA FOR *

* DOCKING: *

*LM3/DAC/10/CEX-ULC *

* (T8,1/250,6) 1 FPS *

* .25 MAG(0), (4MIN) *

V34, P00

V76

MANEUVER/PICTURES OF SIMBAY

50 INITIATE DOCKING

COAS TO OVHD WINDOW

*EXT LTG-DOCK *

SHFT/TRUN +50 *

V41N72 (+000,+320)

CB RR(2)-OPEN, V44

FDAI (180,294,300)

V77

60 CONTACT

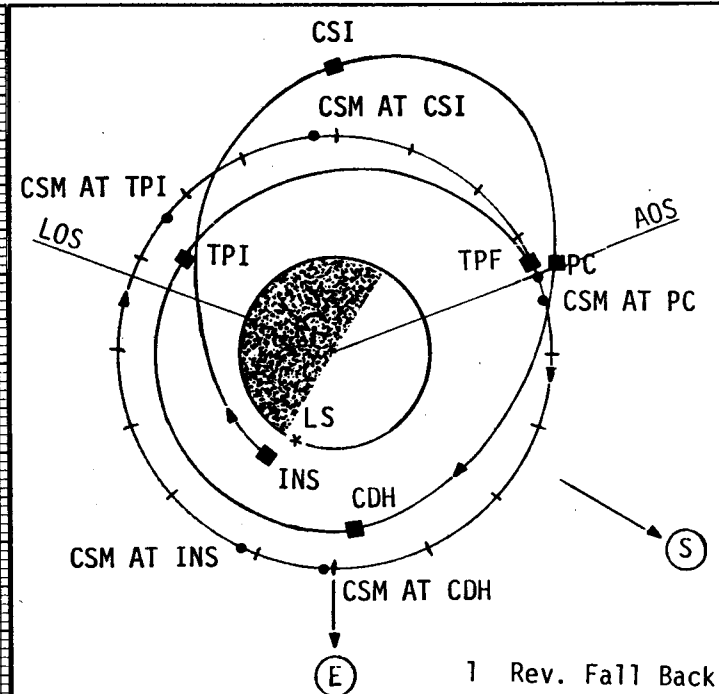
CONFIRM CAPTURE FROM CSM

MODE CONT (BOTH)-OFF

POST DOCKING PROCEDURES

TPI THRU
DOCKING

EVENT	GET TIG
INS	
CSI	
PC	
CDH	
TPI	



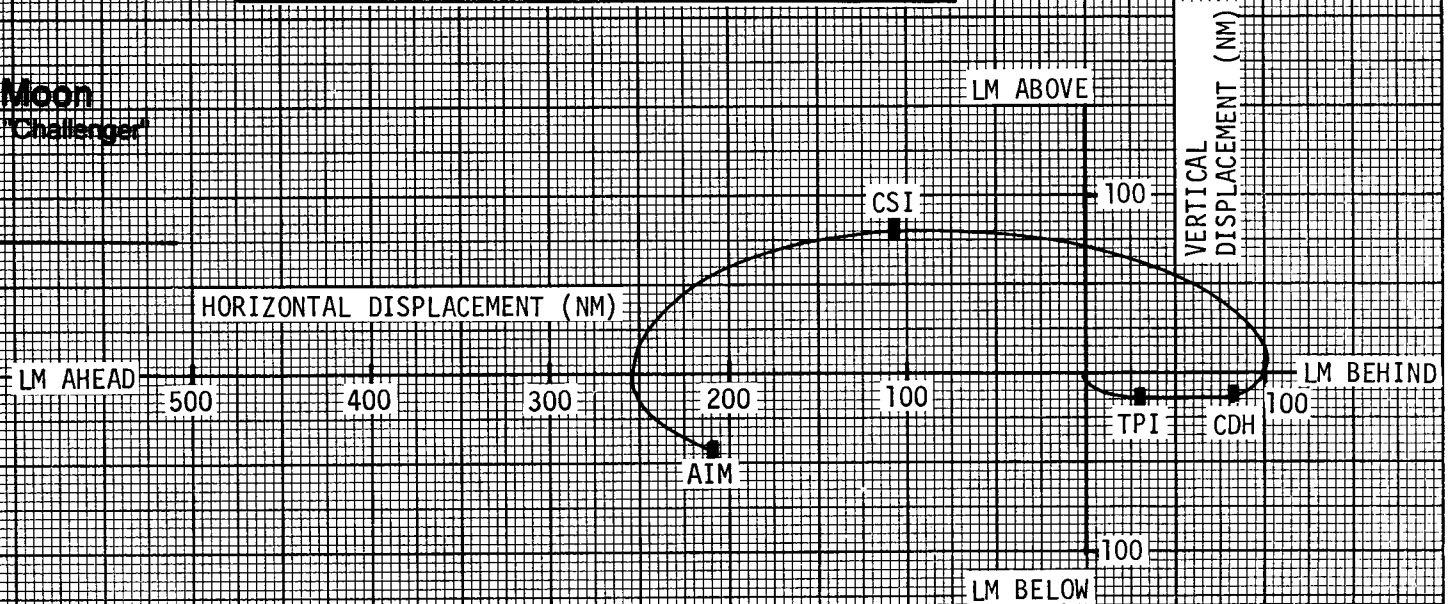
APOLLO 16/17 MISSIONS
INERTIAL AND RELATIVE PLOTS

NO DOI 1 AT PDI 1+12
NO PDI 1+12

Landed on the Moon
aboard the Apollo 17 LM "Challenger"

Gene Cerny

Landed on the Moon
aboard the Apollo 17 LM "Challenger"



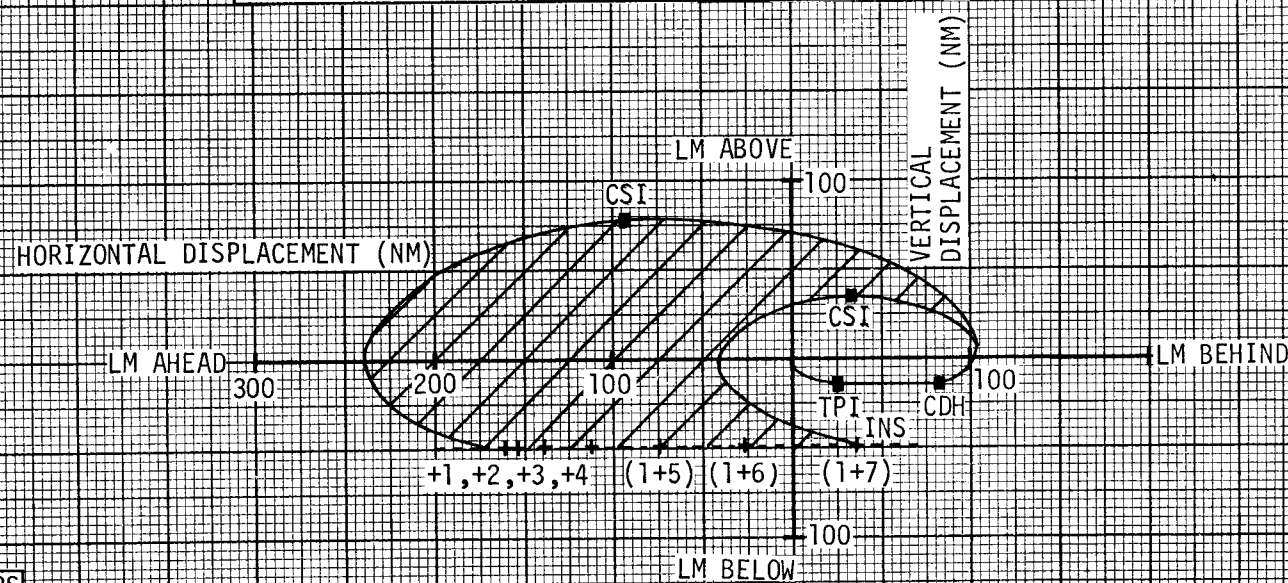
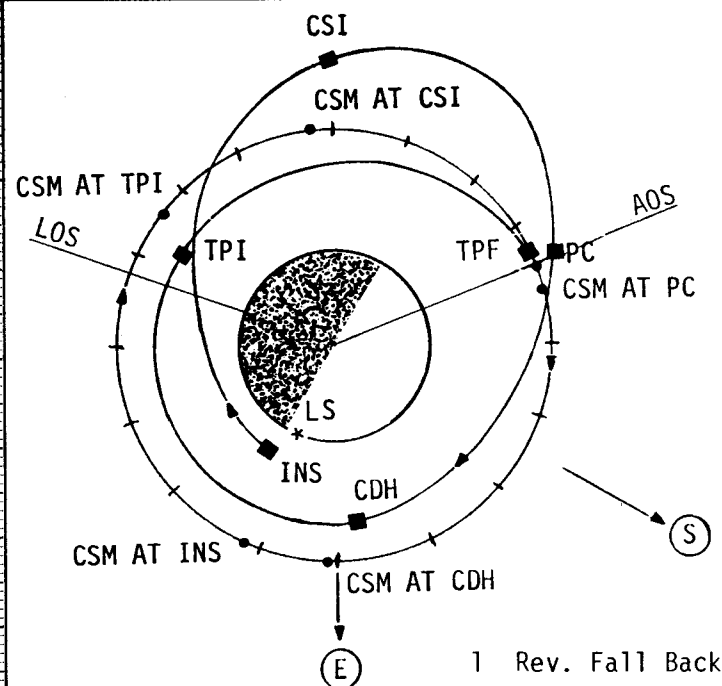
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SEPTEMBER 1, 1972

PAGE A-1

APOLLO 16/17 MISSIONS INERTIAL AND RELATIVE PLOTS

1. $1 \leq \text{PDI} \leq 7$

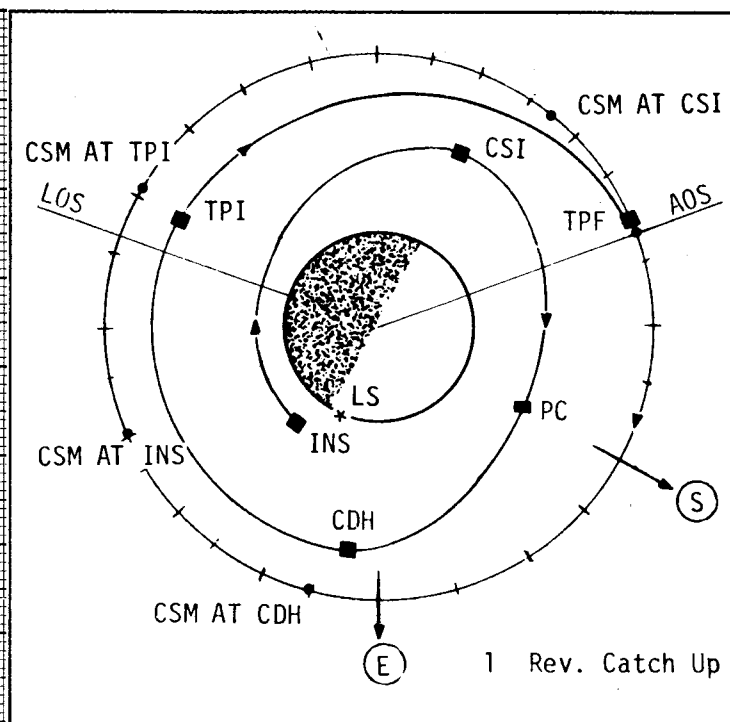
EVENT	GET TIG
INS	
CSI	
PC	
CDH	
TPI	



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PAGE A-2

EVENT	GET TIG
INS	
CSI	
PC	
CDH	
TPI	

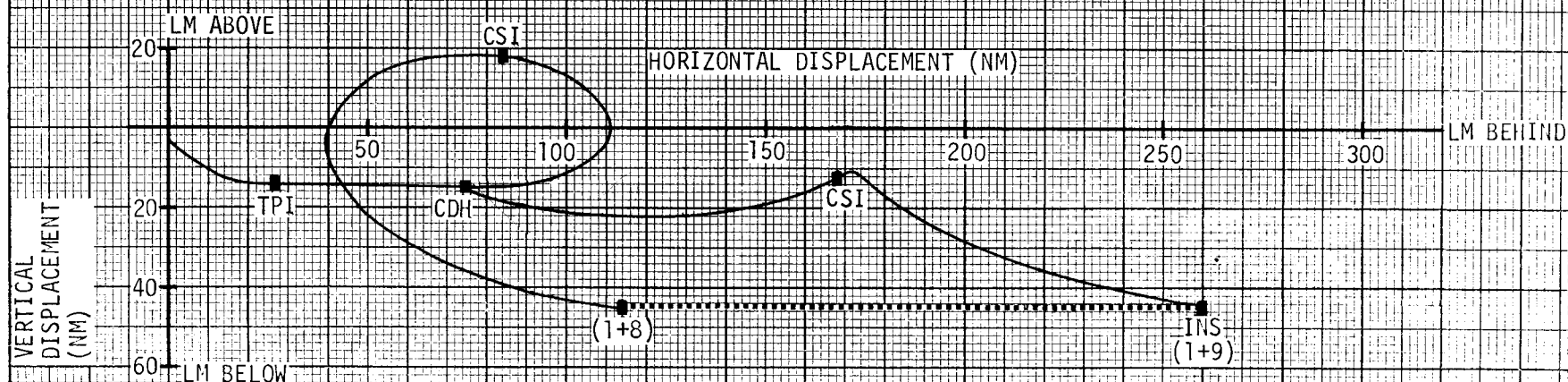


APOLLO 16/17 MISSIONS
INERTIAL AND RELATIVE PLOTS

$8 \leq \text{PDI } 1 \leq 9$

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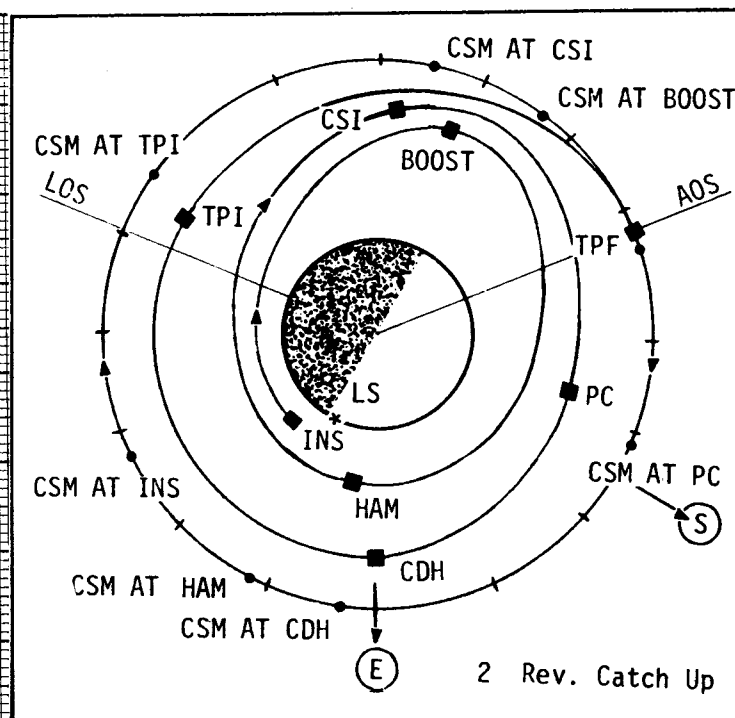


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SEPTEMBER 27, 1972

PAGE A-3

APOLLO 16/17 MISSIONS
INERTIAL AND RELATIVE PLOTS
1. $10 < \text{PDI } 1 \leq 17$
2. T2-2

EVENT	GET TIG
INS	
BOOST	
HAM	
CSI	
PC	
CDH	
TPI	



VERTICAL
DISPLACEMENT (NM)

LM AHEAD

LM ABOVE

HORIZONTAL DISPLACEMENT (NM)

LM BEHIND

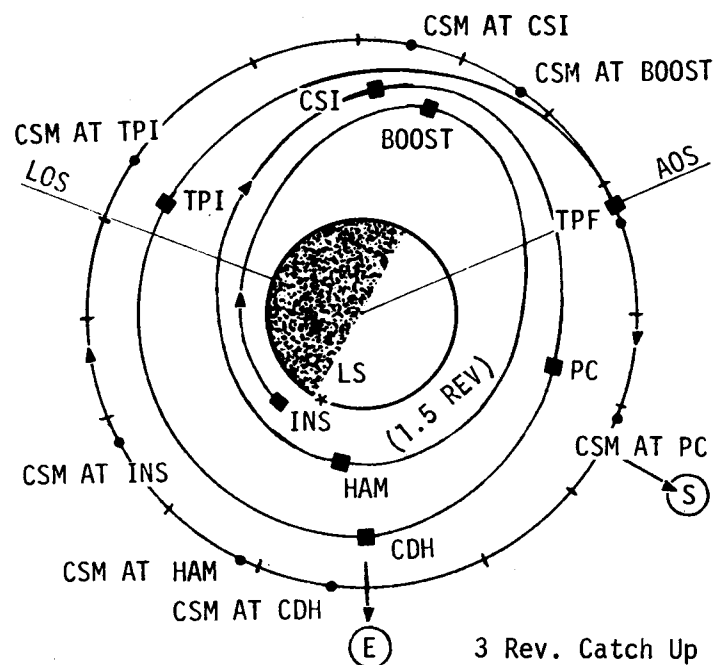
100

LM BELOW

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PAGE A-4

EVENT	GET TIG
INS	
BOOST	
HAM	
CSI	
PC	
CDH	
TPI	



APOLLO 16/17 MISSIONS INERTIAL AND RELATIVE PLOTS

T2-1

PROCEDURES:

BOOST BURN PERFORMED

P00

V82

V76

1 REV COASTING FLIGHT

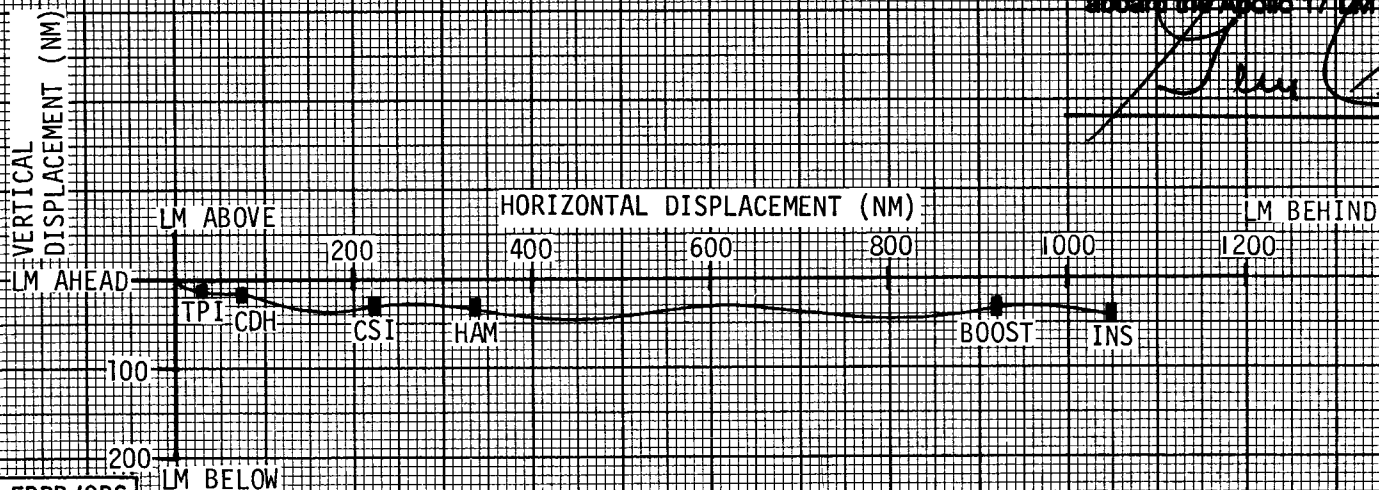
60 MIN PRIOR TO HAM

DO PROCEDURES FOR

BOOST THRU HAM.

ON PAGE 4-5

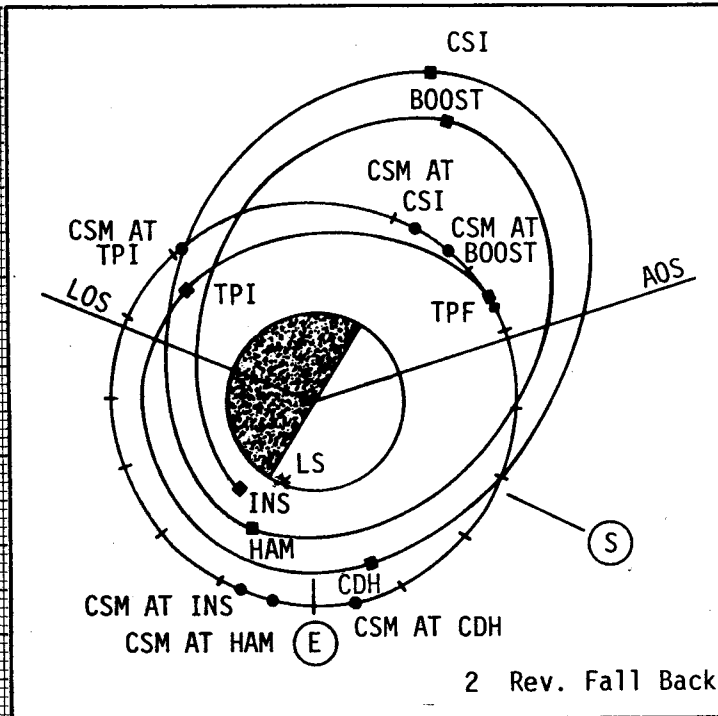
BURN GROUND HAM



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OCTOBER 25, 1972

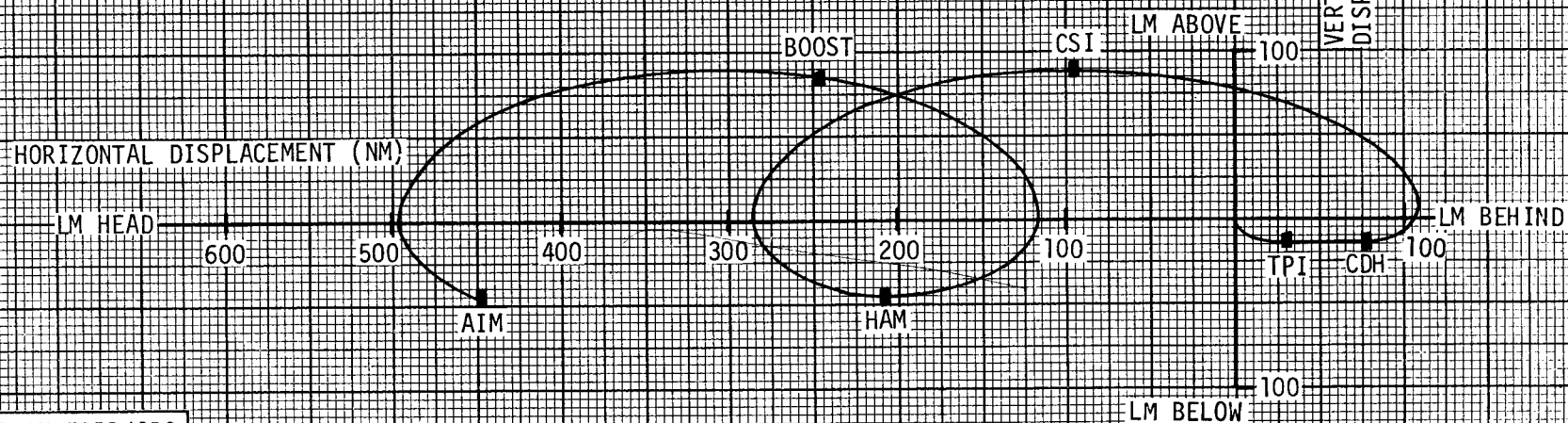
PAGE A-5

EVENT	GET TIG
INS	
BOOST	
HAM	
CSI	
PC	
CDH	
TPI	



APOLLO 16/17 MISSIONS
INERTIAL AND RELATIVE PLOTS

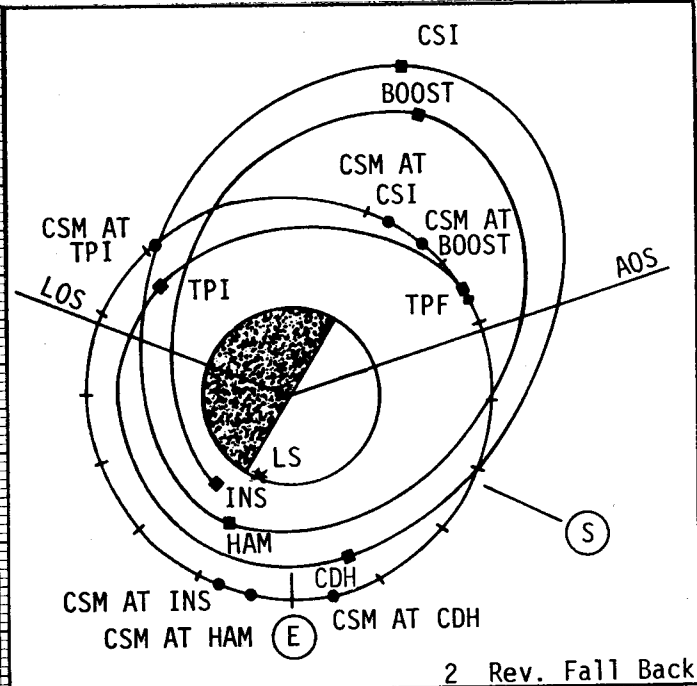
NO DOI 2 AT PDI 2+12
NO PDI 2+12



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PAGE A-6

EVENT	GET TIG
INS	
BOOST	
HAM	
CSI	
PC	
CDH	
TPI	

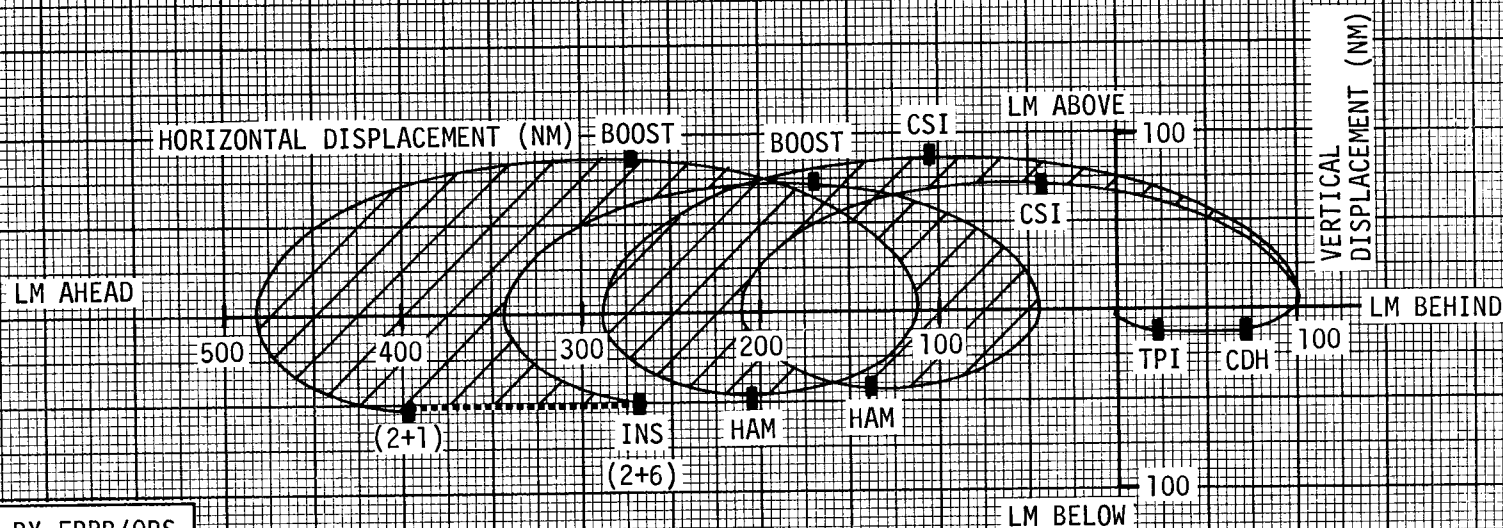


APOLLO 16/17 MISSIONS
INERTIAL AND RELATIVE PLOTS

1. $1 \leq \text{PDI} \leq 6$

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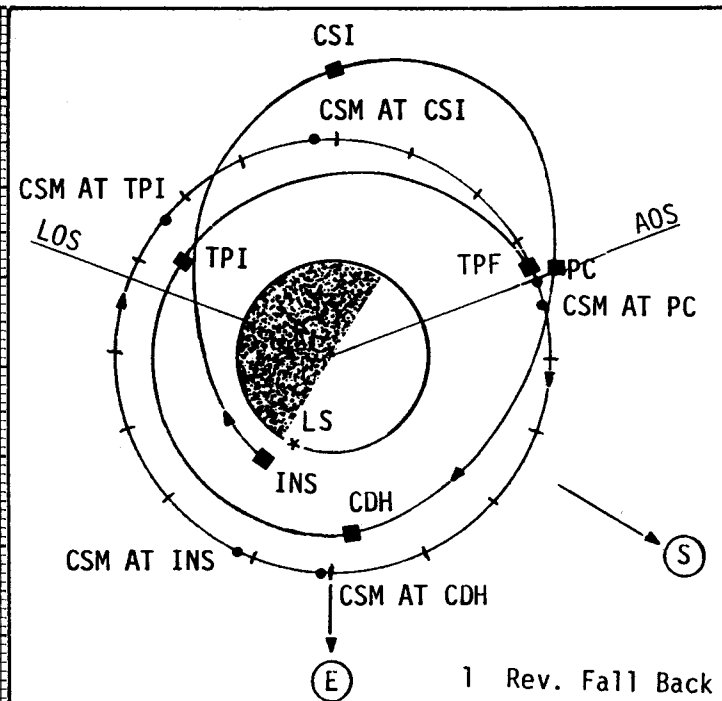
Edwin S. Collins



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SEPTEMBER 1, 1972

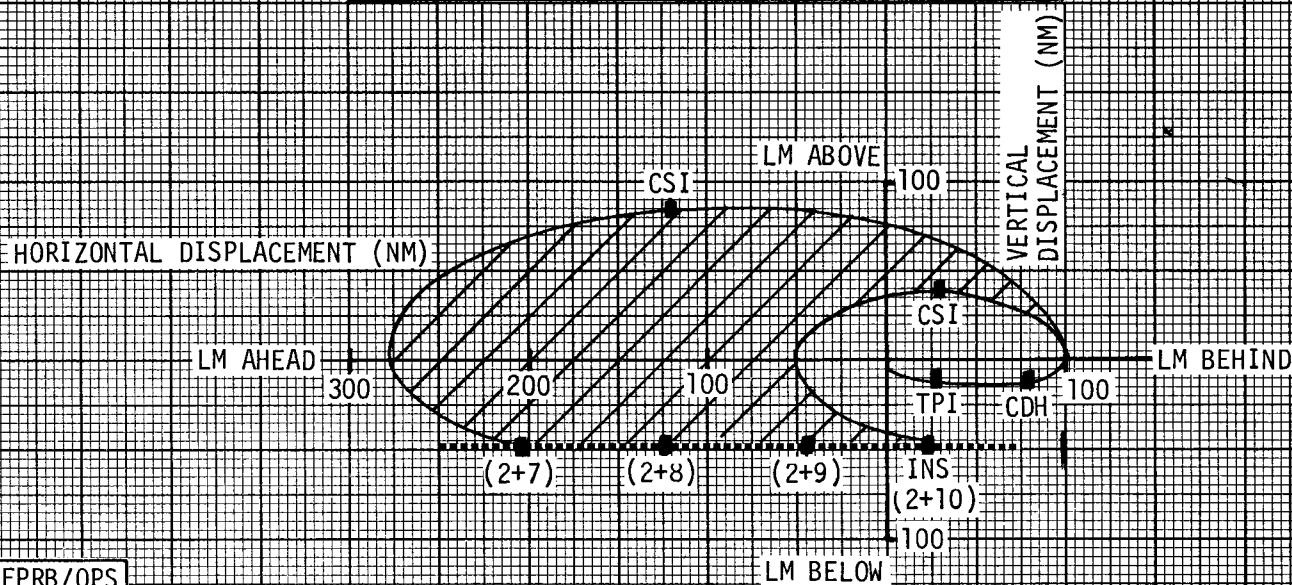
PAGE A-7

EVENT	GET TIG
INS	
CSI	
PC	
CDH	
TPI	



APOLLO 16/17 MISSIONS
INERTIAL AND RELATIVE PLOTS

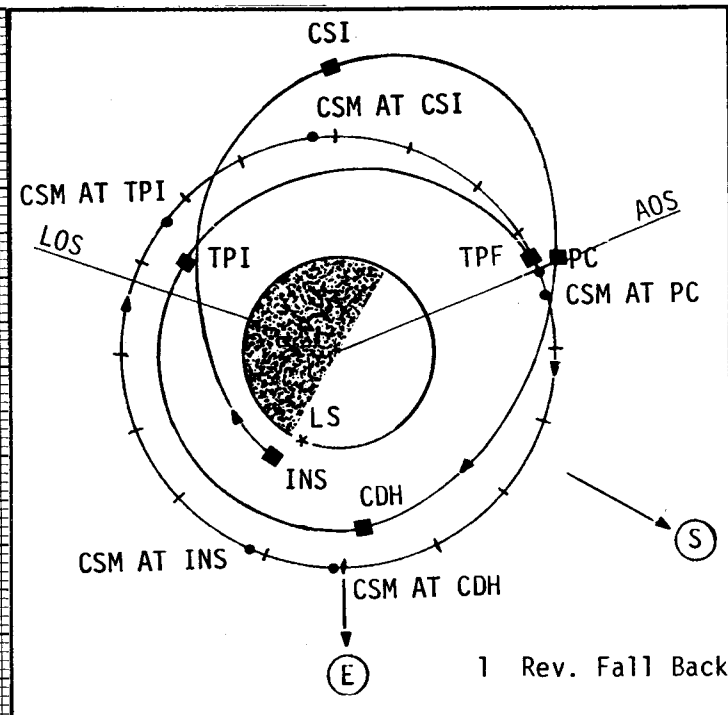
$7 \leq \text{PDI } 2 \leq 10$



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PAGE A-8

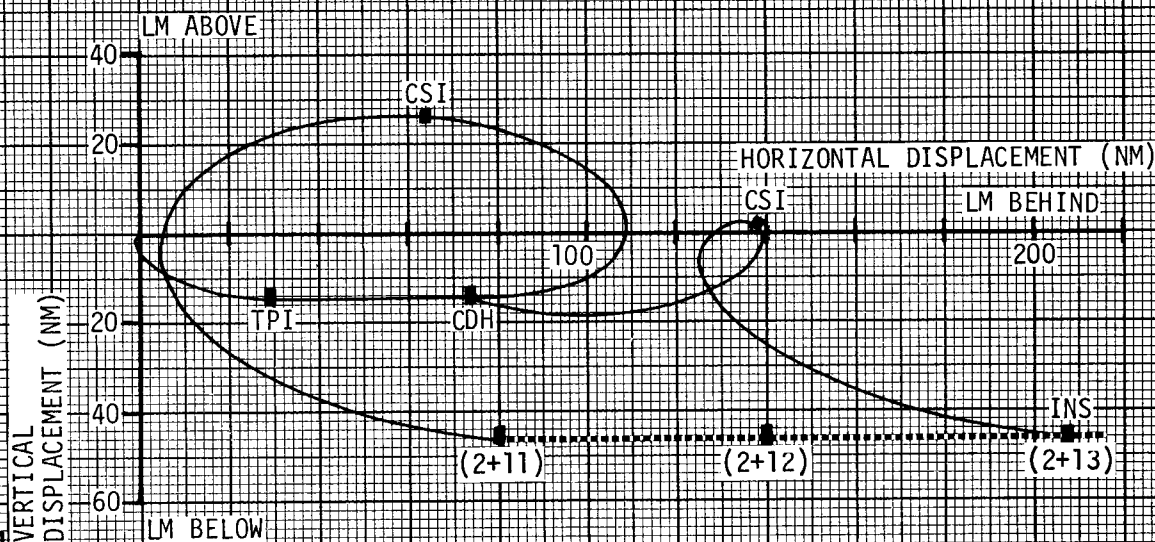
EVENT	GET TIG
INS	
CSI	
PC	
CDH	
TPI	



APOLLO 16/17 MISSIONS
INERTIAL AND RELATIVE PLOTS

11 ≤ PDI 2 ≤ 13

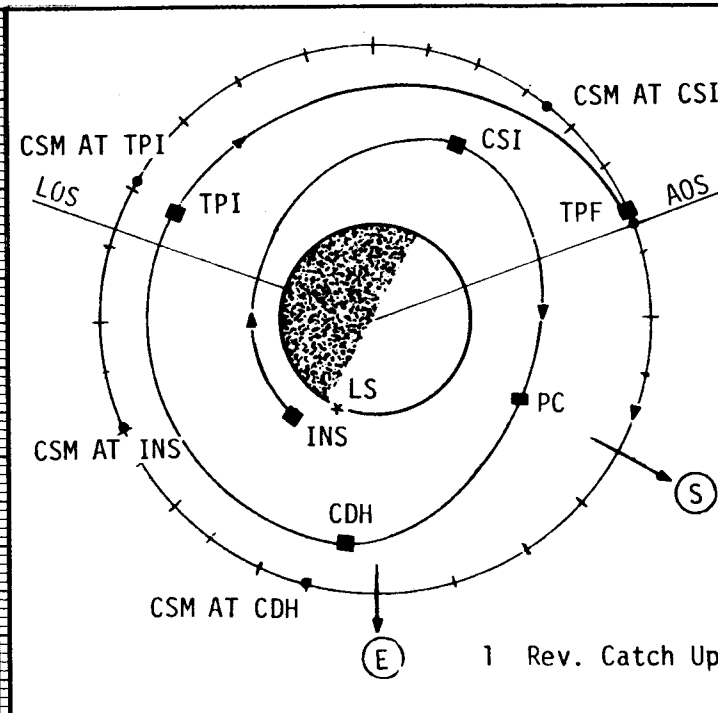
Landed on the Moon
about the Apollo 17 LM "Challenger"



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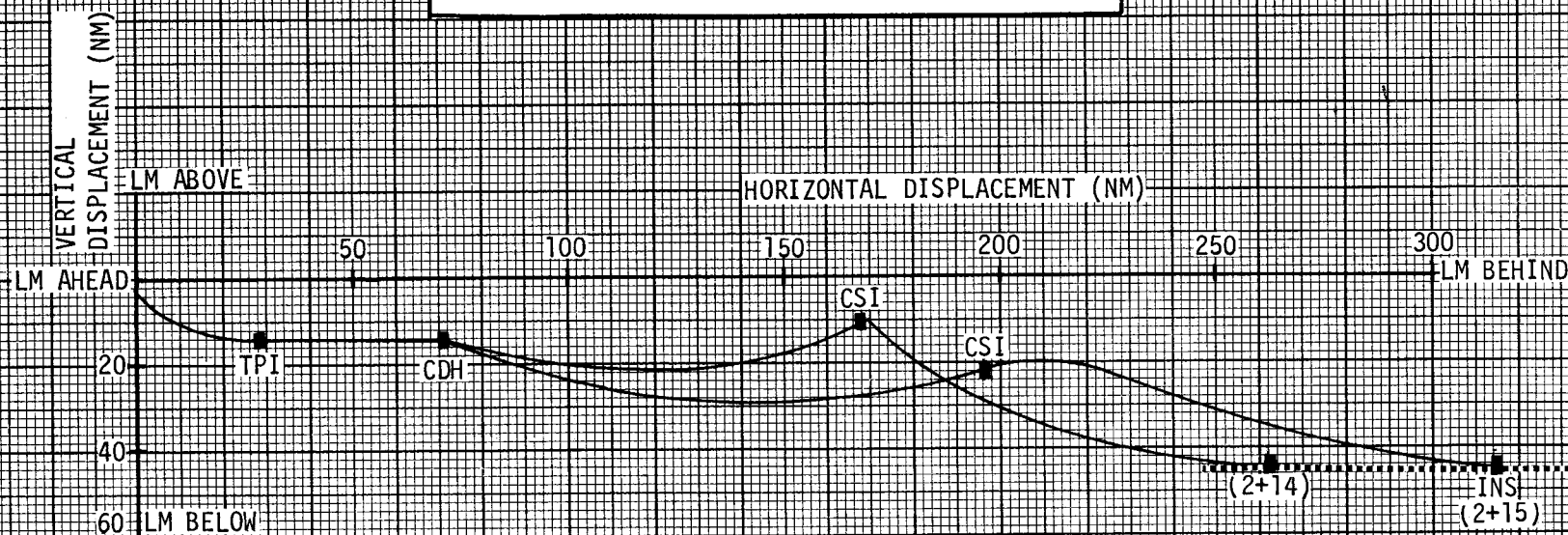
PAGE A-9

EVENT	GET TIG
INS	
CSI	
PC	
CDH	
TPI	



APOLLO 16/17 MISSIONS
INERTIAL AND RELATIVE PLOTS

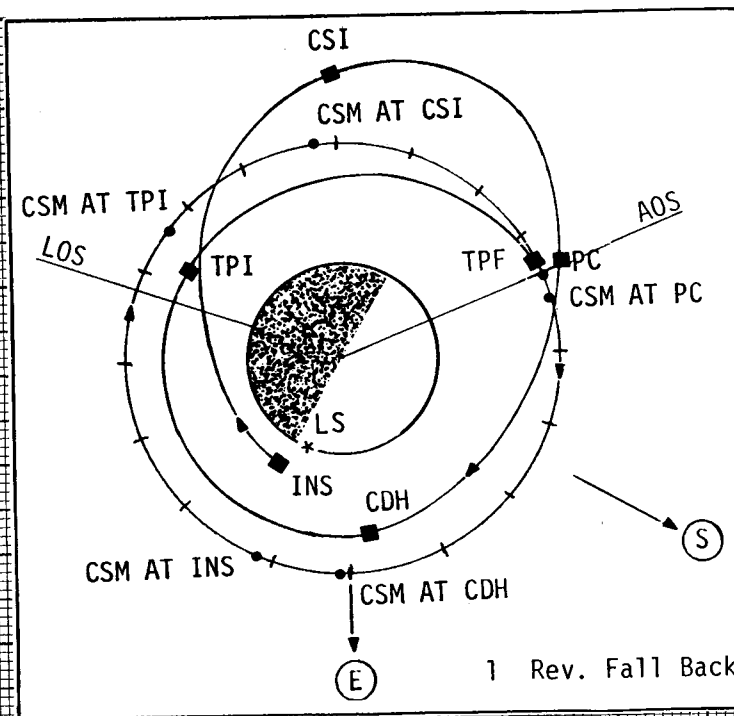
$$14 \leq \text{PDI} \leq 15$$



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PAGE A-10

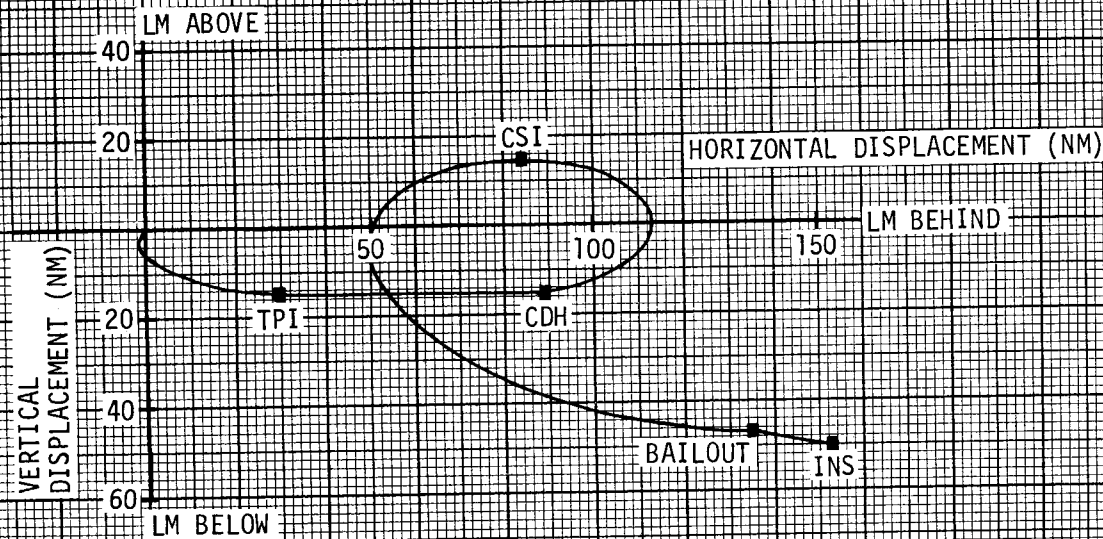
EVENT	GET TIG
INS	
BAILOUT	
CSI	
PC	
CDH	
TPI	



APOLLO 16/17 MISSIONS
INERTIAL AND RELATIVE PLOTS

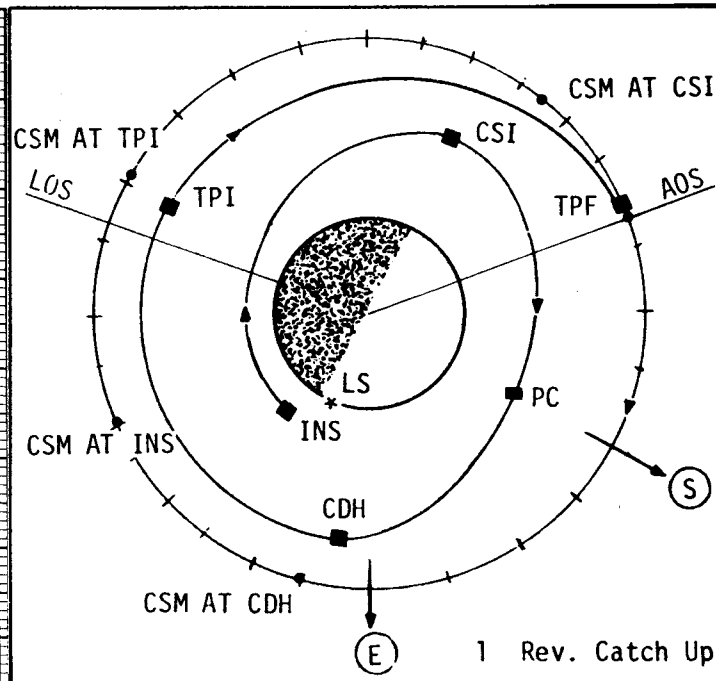
BAILOUT

Landed on the Moon
aboard the Apollo 17 LM "Challenger"



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DECEMBER 25, 1971

EVENT	GET TIG
INS	
CSI	
PC	
CDH	
TPI	



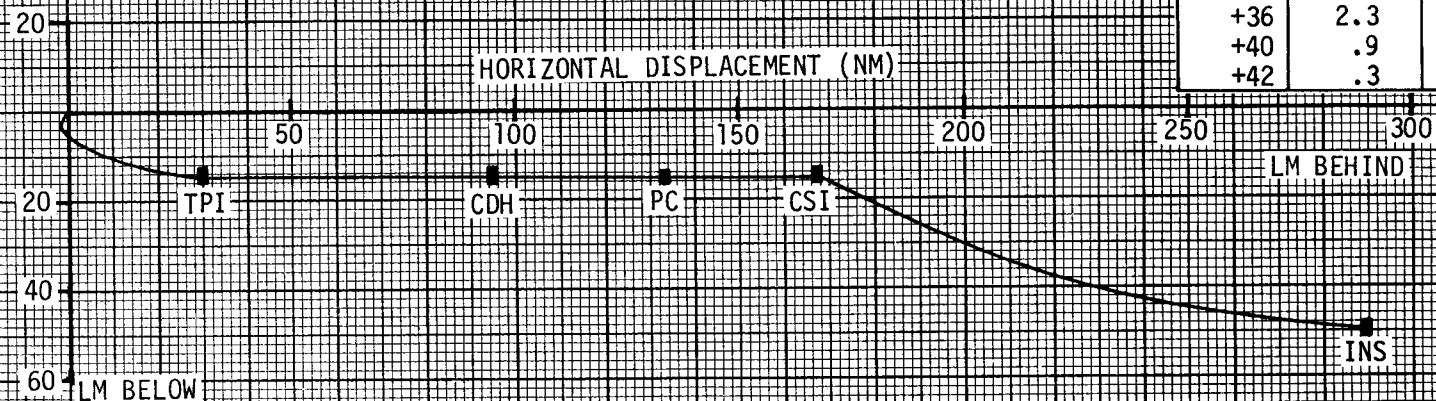
APOLLO 16/17 MISSIONS INERTIAL AND RELATIVE PLOTS

NOMINAL COELLIPTIC RNDZ

ΔT	RANGE	RDOT
INS	285.6	-456.6
-48	281.1	-452.3
-44	263.6	-429.7
-40	247.3	-399.1
-36	232.2	-362.0
-32	218.7	-320.2
-28	206.9	-275.8
-24	196.9	-230.9
-20	188.7	-188.0
-16	182.0	-149.3
-12	176.8	-116.8
-8	172.7	-92.0
-4	169.4	-76.0
CSI	166.6	-69.4
CDH	95.7	-122.2
TPI	33.5	-110.2
+4	27.8	-129.3
+8	22.9	-120.4
+12	18.4	-108.8
+16	14.3	-95.2
+20	10.9	-80.4
+24	8.0	-65.7
+28	5.6	-52.5
+32	3.8	-42.0
+36	2.3	-35.3
+40	.9	-32.4
+42	.3	-31.9

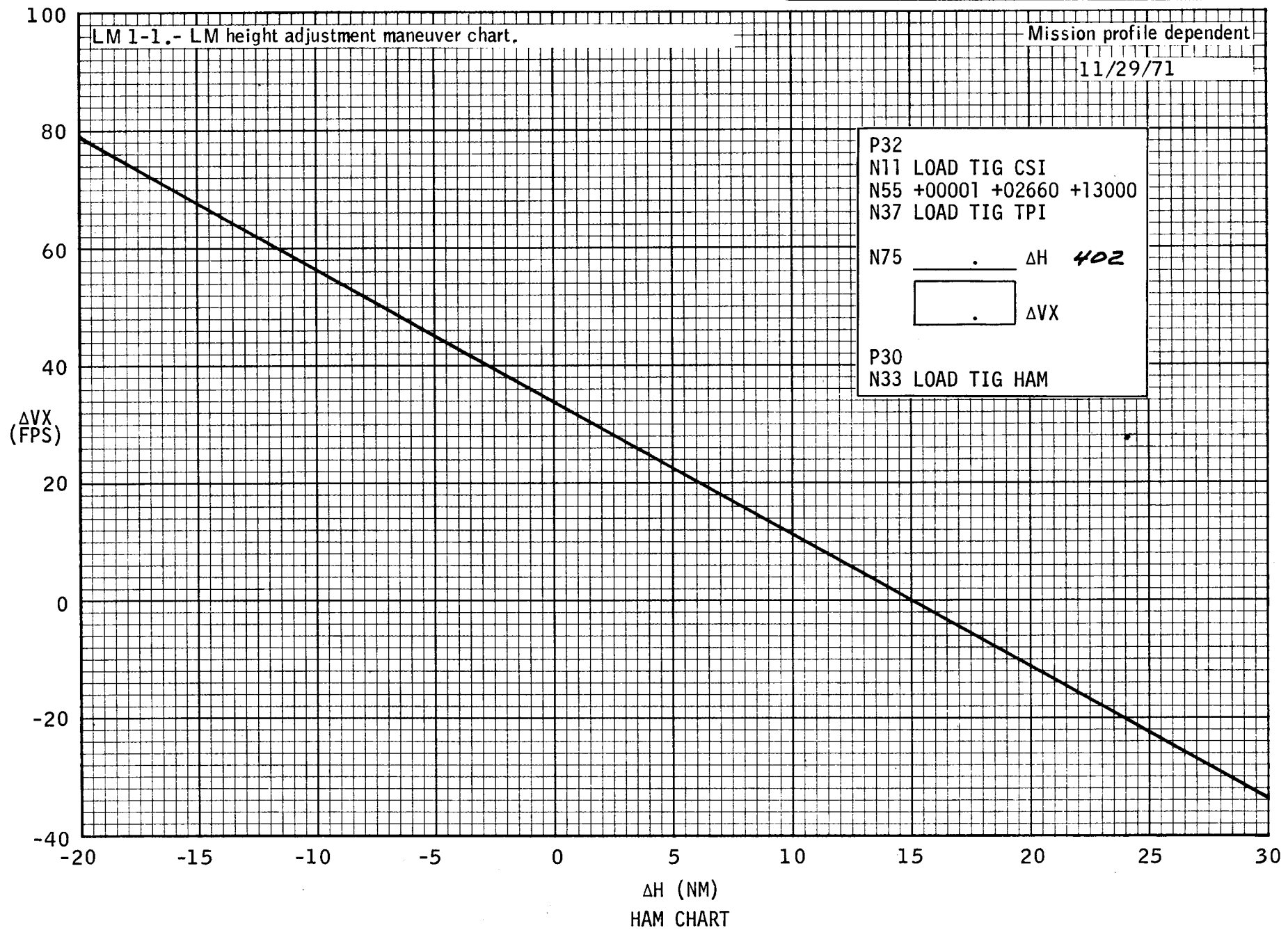
Landed on the Moon
aboard the Apollo 17 LM - Challenger

VERTICAL
DISPLACEMENT
(NM)



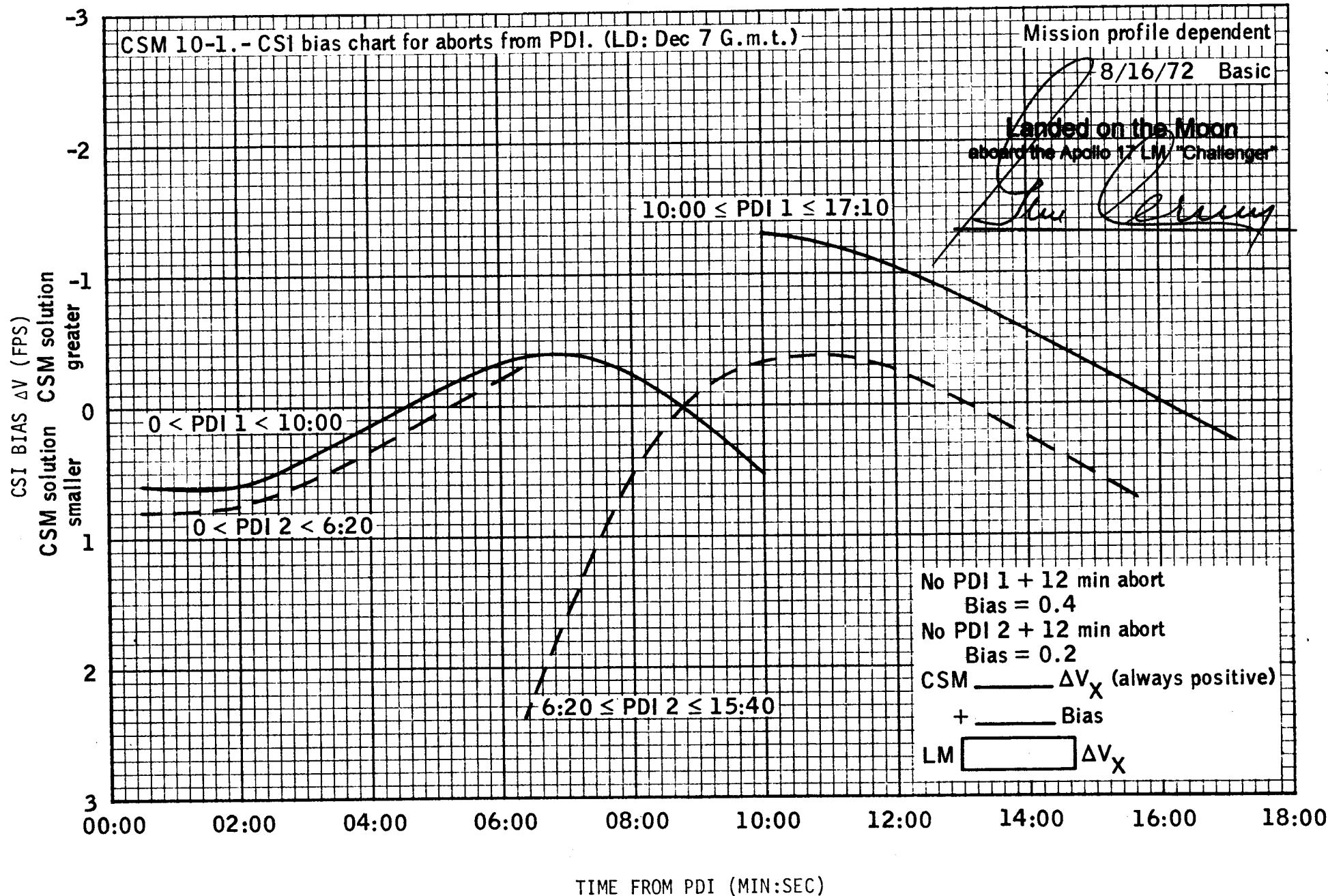
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DECEMBER 25, 1971

RELATIVE TRA
NOM COEL RNDZ



CSI BIAS CHART

DATE 9/8/72



CSI BIAS
DELTA V

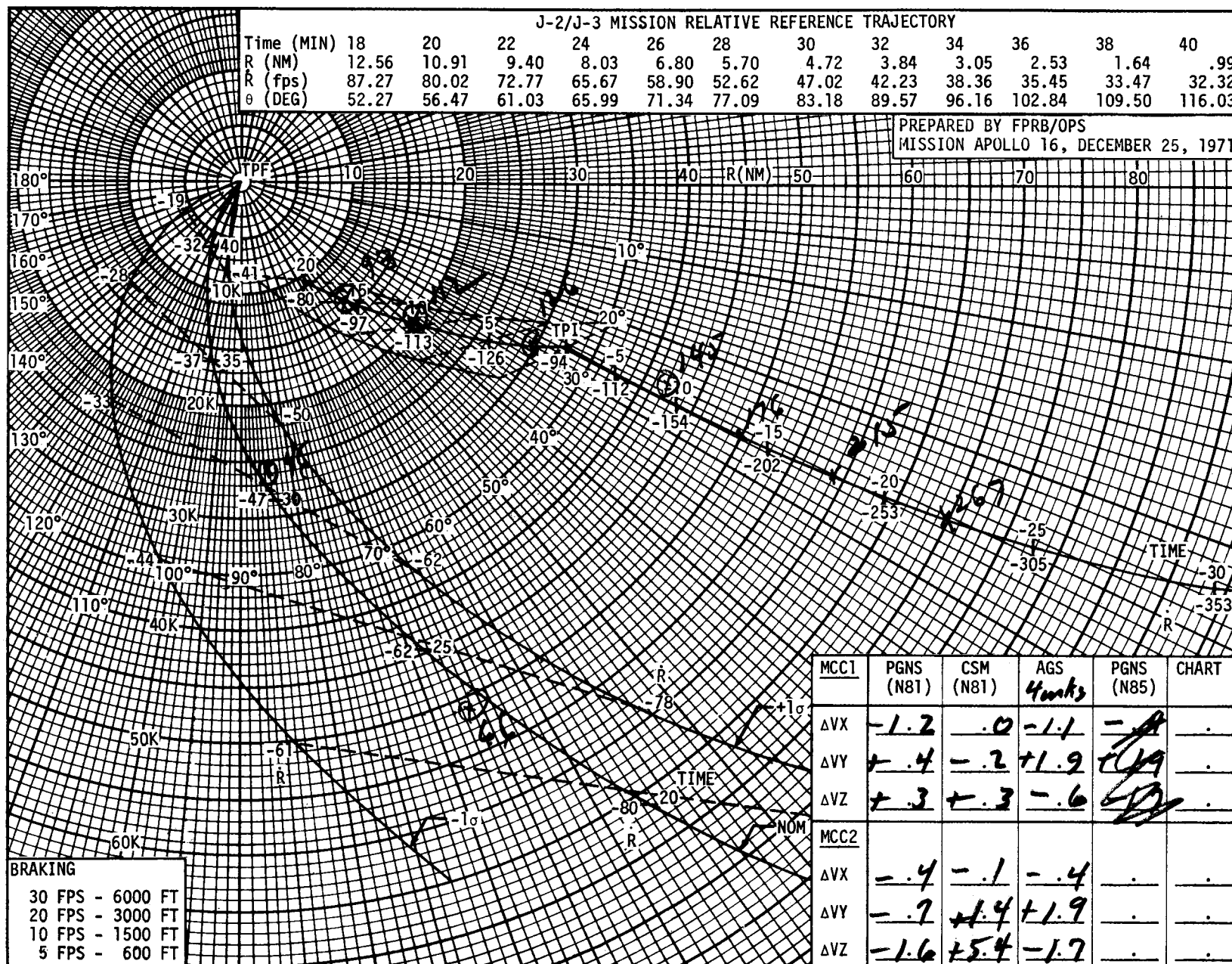
PAGE 4-23

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CSI BIAS
DELTA V

Landed on the Moon
aboard the Apollo 17 LM "Challenger"

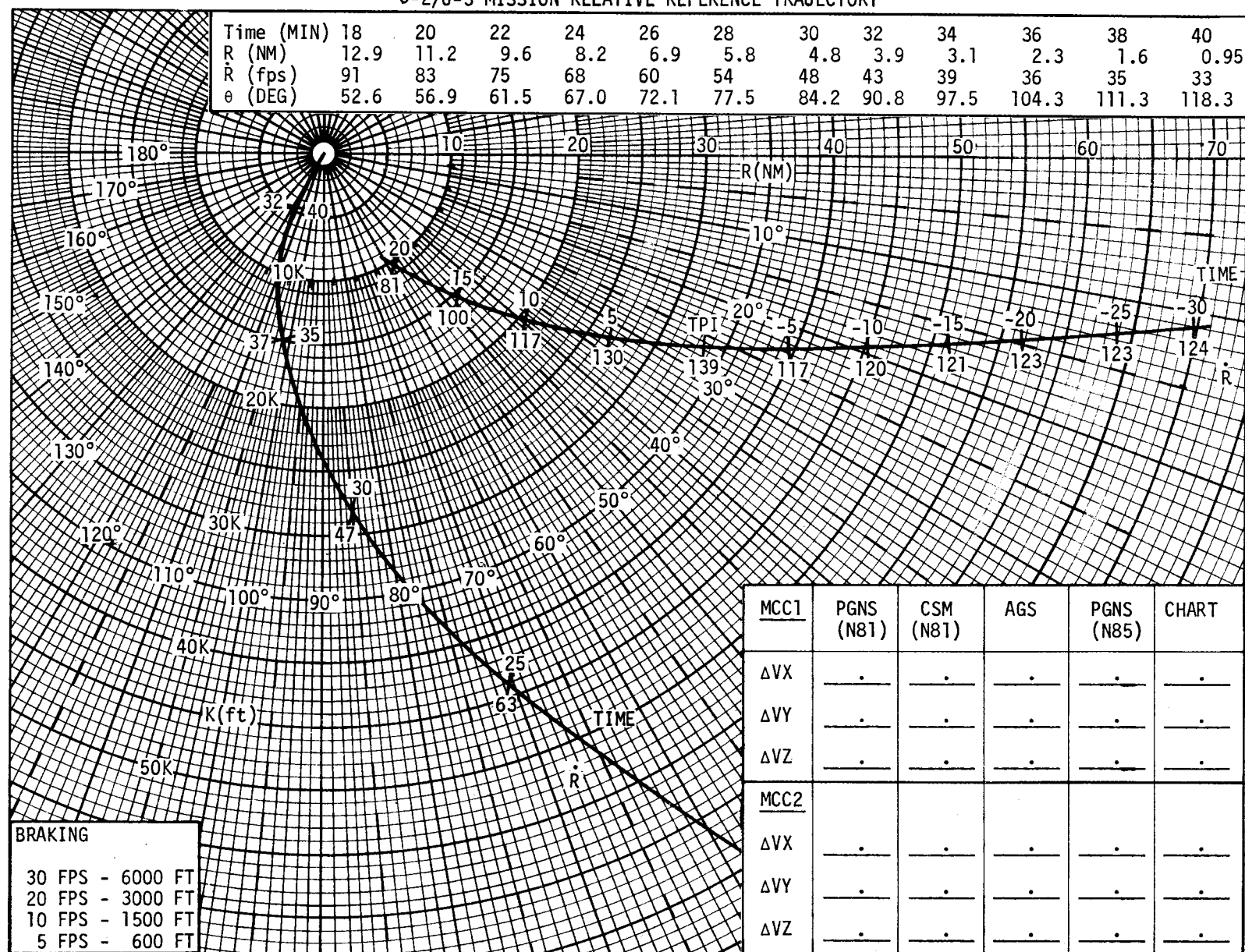
Gene Cerny



THIS IS THE END

NOT THE BEGINNING

J-2/J-3 MISSION RELATIVE REFERENCE TRAJECTORY



PREPARED BY FPRB/OPS
MISSION APOLLO 16, DECEMBER 25, 1971

DATE 9/8/72

TPF REL TRAJ
NOM COEL RNDZ